

Language as Visible Signal: Toward an Image-Native Language Model in Continuous State and Pixel Space

Imagized Language Model Project

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Abstract

Most language models make a finite symbolic encoding the primary interface to language. That interface is effective for modern encoded text but incomplete for historical, handwritten, regional, damaged, and unencoded writing, where visual form is part of the evidence. We study a stricter image-native contract: the deployed model receives writing pixels, maintains continuous visual states, predicts continuous visual futures, emits a raster answer, and rereads the visible pixels it generated. OCR and coded transcription remain optional post-hoc views rather than the model’s language substrate.

We implement this contract through staged, falsifiable experiments. V34 is a codebook-free convolutional codec with 7.42 million parameters for 32×32 writing patches. It qualifies on held-font and held-family development and sealed audits, reaching 0.9978 rendered ink F1 and 0.9981 historical ink F1 on the sealed split with 2.38 GiB peak allocated memory. Reconstruction is only an interface result. V35 combines the frozen codec with a residual interface adapter, a 12-layer PIXAR-initialized causal transformer, deterministic anchor and conditional rectified-flow writers, a stop head, and a strict decode-threshold-re-encode feedback loop. PIXAR is an external causal foundation, not a project contribution; V35 is transfer-based rather than from-scratch pretraining.

The 129.09M-parameter V35 student completes 22,000 finite BF16 updates in 2.77 hours on one RTX 4090 with 2.899 GiB peak allocated memory. Its packaged runtime accepts only pixels and a visual-position mask and calls no tokenizer, Unicode or character IDs, OCR, lookup table, retrieval system, codebook, or teacher. Nevertheless, the fixed development audit rejects the mechanism. Autonomous copy accuracy is 0.3125%, retaining only 0.491% of its target-through-codec OCR ceiling; correct prompts improve over shuffled prompts by only 0.117 percentage point. Instruction accuracy is 0.1116% and is 0.445 point worse than the shuffled-prompt condition. The visual-causal and semantic-raster routes pass only 5/12 and 5/9 gates, respectively. Generated pixels are finite, nonblank, and prompt-responsive, but contain corrupted glyph-like marks rather than correctly bound answers. The decision is **not-qualified**, and the sealed split remains unopened. Thus V35 establishes a complete one-GPU raster-input/raster-output experimental runtime, not useful autonomous language, parity with text LLMs, or superior compute efficiency. The result localizes the next problem to answer-level semantic planning and prompt-to-output binding before local raster realization.

1 Introduction: Writing as Language

1.1 Pain point

Human readers encounter language as marks on pages, screens, stone, bone, metal, and moving hands. Those marks vary with writer, era, medium, damage, typography, and reading direction. A

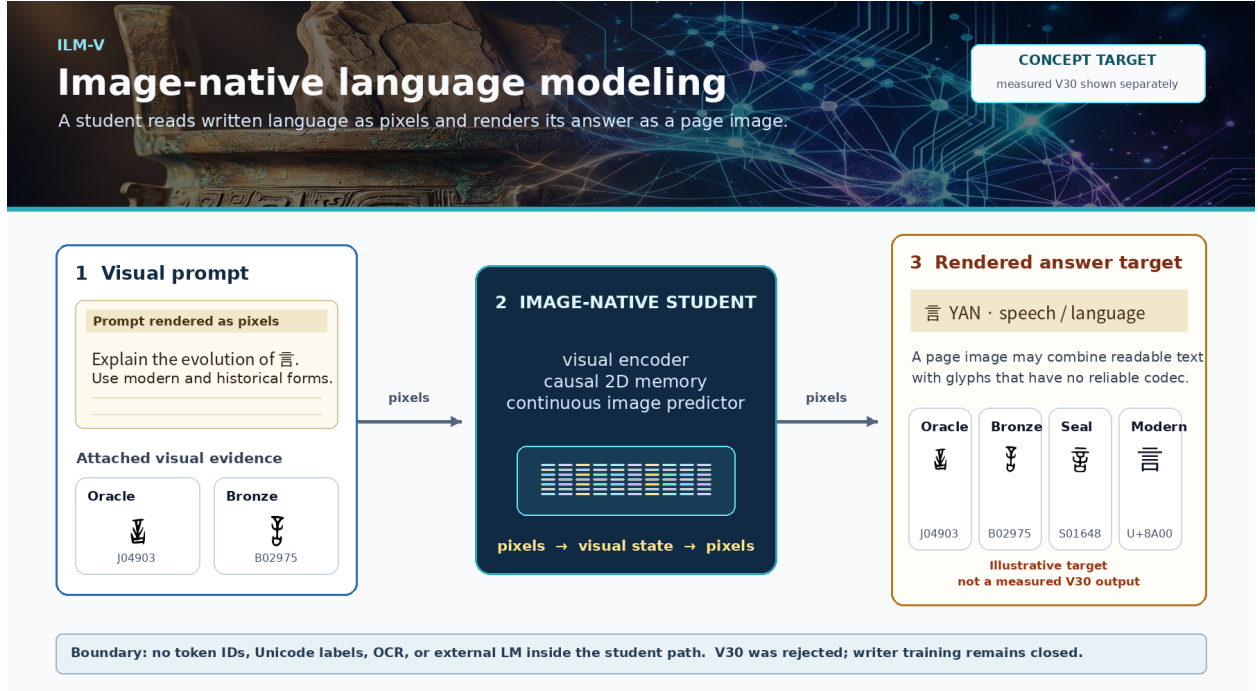


Figure 1: Image-native language modeling design target. A restrained AgInTi background is overlaid with deterministic typography and real local ziyuan glyph exemplars for YAN (U+8A00). The right panel is an illustrative answer-page target, not a measured model output; the latest measured result is the rejected V35 experiment in Figure 24.

finite token or Unicode interface is therefore not a generally accessible inventory of written language: it is an extremely useful codec for a documented subset. Historical and regional forms may have no code point, two visually different forms may share one code point, and an OCR transcript may erase the form whose evolution is the question.

Current language models are extremely capable, but their central language state is normally a sequence of symbolic tokens. Multimodal models can accept images, and modern APIs expose image generation [4, 5, 6]. These systems are important baselines, but attaching vision to a token-centered model does not make visible writing the predictive language substrate. An OCR system similarly solves a useful projection from image to available codes; it does not model all of the visual language that was projected away.

1.2 Goal and scope

The Imagized Language Model (ILM) instead treats written form as language:

$$\text{image input} \rightarrow \text{visual language state} \rightarrow \text{image output}.$$

The output should itself be a readable answer image. For a query such as “explain the evolution of character yan”, the system should produce a page containing modern explanation and attested oracle, bronze, seal, and modern forms, including forms unavailable in computer fonts. Typed input is rasterized before the core. A photographed page can be transcribed, but the same model should also continue, translate, explain, compare, and generate writing. Its answer is an image first; coded transcription is an optional downstream view.

The canonical input/output can be written as a Visual Language Stream,

$$X \in [0, 1]^{T \times D \times H \times W \times C}, \quad p_\theta(X_{t+1:t+k} \mid X_{\leq t}),$$

where T is reading sequence or visual time, D is optional geometric depth, H, W are spatial extent, and C is the sensor channel. A flat prompt has $T = 1, D = 1$; an ordered line or book has $T > 1, D = 1$; a 3D glyph string has $D > 1$; and a 3D character movie uses depth and time. Collapsed axes have length one, so these are shapes of one continuous field rather than separate vocabularies. The present proof fixes $D = 1$ and short T ; depth and motion must earn separate causal and quality evidence after the 2D loop passes.

Prompt following partitions this stream into observed prompt frames and generated answer frames,

$$X_{\text{prompt}} \in [0, 1]^{T_p \times D \times H \times W \times C} \longrightarrow Y_{\text{answer}} \in [0, 1]^{T_a \times D \times H \times W \times C}.$$

A typed question is deterministically rendered into prompt frames; a scan or inscription enters natively. $T_a = 1$ is an answer page and $T_a > 1$ is an ordered text-image stream or movie. A language claim requires correct held-out changes in the generated answer when the prompt meaning changes. Input reconstruction, OCR accuracy, glyph classification, or attractive writing alone is insufficient.

The design is inspired by foveated visual reading, without claiming a biological simulation. Its broader hypothesis is sensory: the same perception–memory–prediction–motor–self-perception loop might later operate on speech waveforms or time-frequency fields. This paper tests visible writing only.

1.3 Contributions

The contribution is not the slogan *text as pixels*, nor a priority claim over earlier pixel language models. It is an implemented sequence of interfaces and controls that makes a stronger image-native language claim falsifiable:

1. A strict deployed-student boundary: after optional typed-prompt rasterization, model methods receive pixels and masks and return pixels. Token and Unicode IDs, strings, OCR, finite visual codebooks, glyph lookup, retrieval, and runtime teacher calls are excluded and audited.
2. A staged experimental record separating visual identity, continuous actuation, relational binding, generated-pixel rereading, natural continuation, and autonomous writing. Positive bounded mechanisms and negative natural-language results are retained rather than collapsed into a single capability claim.
3. V34, a qualified 7.42M-parameter continuous writing codec that preserves held-font modern Chinese and held-family historical forms without quantization. Its sealed reconstruction result establishes an interface, not language understanding.
4. V35, a 129.09M-parameter causal raster student combining that frozen codec with an explicitly credited PIXAR initialization, anchor and rectified-flow writers, and a visible decode–threshold–re-encode recurrent boundary. It trains and runs on one RTX 4090 without a symbolic deployed path.
5. A preregistered development audit with correct, shuffled, blank, partial, paraphrased, and copy-counterfactual prompts; target-through-codec ceilings; writer selection; direct raster galleries; immutable hashes; and a sealed split that may open only after qualification. These controls reject V35’s prompt-to-answer binding despite nonblank prompt-responsive pixels.

6. A rights-aware Visual Word-Origin Book target and Visual Language Stream interface for modern and historical writing. Historical answer generation, page-scale memory, depth, motion, speech, and broad multilingual capability remain prospective until they pass their own causal and quality controls.

2 Prior Art and Gap

Pixel language modeling is established prior art. PIXEL treats rendered text as patches for masked representation learning and cross-script transfer [10]. Rendering strategy is itself consequential: character-bigram rendering enabled a 22M PIXEL variant to match its 86M reference on the studied tasks while exposing patch-frequency anisotropy [11]. PIXEL-M4 extends encoder pretraining to English, Hindi, Ukrainian, and Simplified Chinese and reports multilingual semantic and syntactic transfer [12]. These are encoder-side results rather than autonomous raster answering.

PIXAR directly demonstrates decoder-only autoregressive language modeling from and to binary rendered patches [13]. MIXAR scales that family across eight languages and multiple scripts [14], while PixelGPT studies autoregressive pretraining across pixels and text [15]. Accordingly, we make no priority claim for pixel language modeling or pixel generation. V35 deliberately reuses PIXAR’s released causal weights so that one-GPU experiments test a new interface and binding hypothesis instead of attempting to reproduce industrial pretraining. The audited PIXAR source revision is MIT; the separately distributed weight archive contains no stated weight license, so the derived standalone artifact is local-research-only and is not redistribution-authorized.

OCR-free document systems such as Donut and Pix2Struct map document images to task outputs without a separate OCR engine [8, 9]. Token-free models such as CANINE, ByT5, MEGABYTE, BLT, and Scratchpad Patching remove subwords or allocate compute dynamically, but retain symbolic character or byte streams [16, 17, 18, 19, 20]. These are strong practical baselines. They differ from the stricter deployed ILM boundary in which visible writing is both the recurrent input and primary output.

Continuous image generation supplies a second foundation. MAR models each continuous image latent with a diffusion loss rather than vector quantization [28]; Continuous VAR derives non-quantized autoregression from strictly proper scoring rules [38]; and NextStep-1 pairs a large autoregressive backbone with a flow-matching image head [29]. Representation Autoencoders show that reconstructive fidelity alone does not guarantee a representation well suited to generation [30]. TextLDM reaches a parallel conclusion for continuous text latents: alignment to a pretrained language representation is critical, while reconstruction alone is insufficient [43]. V35’s negative result is consistent with that warning: an almost lossless local glyph codec does not by itself supply an answer-level semantic plan.

The scoped gap tested here is therefore not *pixels instead of tokens*. It is the combination of a continuous non-quantized writing representation, a causal visual state, direct generated raster output, and a recurrent boundary in which the model must reread its own thresholded visible pixels. Correct, shuffled, blank, partial, paraphrased, and counterfactual prompts distinguish semantic binding from reconstruction, frequency, or merely changing pixels. Historical forms add a provenance constraint: generated marks must never be presented as attested source glyphs. External components are allowed when they work well, but their origin, license, training-time role, and presence or absence at runtime must remain explicit.

Retinal Flow Language Model

Read visible fixations, predict a visual distribution, write ink, then read the model's own ink.

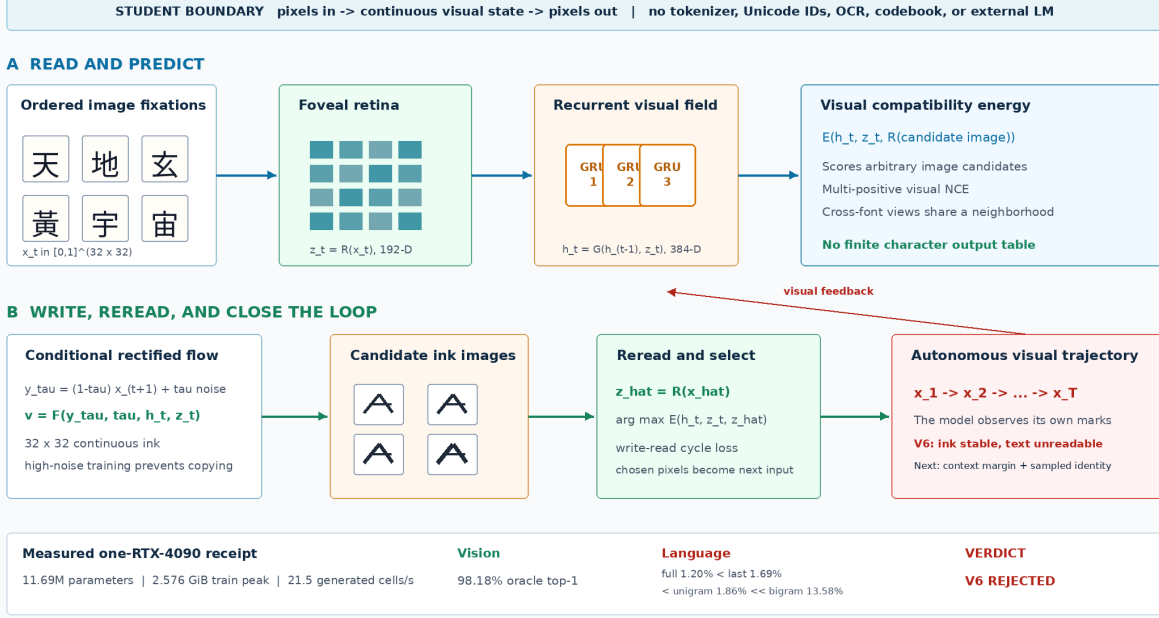


Figure 2: The implemented Retinal Flow Language Model. Training joins visual next-state energy, conditional rectified flow, a write–read cycle, V6 model-induced visual rollouts, and V7 calibrated context and sampled-image identity. Autonomous inference samples candidate ink images, rereads them, selects by visual energy, and feeds the selected pixels back.

3 Architecture

Figure 2 gives the strict visual contract. For fixation images $x_t \in [0, 1]^{H \times W}$, RFLM factors a visual sequence as

$$p_{\Theta}(x_{1:T}) = p(x_1) \prod_{t=1}^{T-1} p_{\Theta}(x_{t+1} | x_{\leq t}).$$

The random variables are continuous image fields rather than entries in a linguistic or visual vocabulary.

Foveal retina. A convolutional reader maps each 32×32 image to a normalized 192-dimensional visual state:

$$z_t = R_{\theta}(x_t) / \|R_{\theta}(x_t)\|_2.$$

Independent font renderings are aligned with visual contrastive, invariance, and variance objectives. The target retina $R_{\bar{\theta}}$ is an exponential moving average of the online retina. Font coverage is checked from font cmap tables so missing-glyph boxes cannot become a spurious class.

Recurrent visual field and energy. A three-layer GRU integrates the fixation history,

$$h_t = G_{\phi}(h_{t-1}, z_t), \quad h_t \in \mathbb{R}^{384}.$$

An energy function scores any candidate image c_j after rereading it:

$$s_{t,j} = E_\psi(h_t, z_t, R_{\bar{\theta}}(c_j)).$$

Multi-positive visual NCE admits multiple near-identical image views as valid targets. Unlike a softmax language head, this energy is not tied to a finite character inventory.

Conditional ink flow. For clean target x_{t+1} , noise ϵ , and $\tau \in [0, 1]$, define

$$y_\tau = (1 - \tau)x_{t+1} + \tau\epsilon, \quad u_\tau = \epsilon - x_{t+1}.$$

The pixel writer predicts the path velocity

$$\hat{u}_\omega = F_\omega(y_\tau, \tau, h_t, z_t), \quad \mathcal{L}_{\text{flow}} = \|\hat{u}_\omega - u_\tau\|_2^2.$$

Training emphasizes high-noise times, where target pixels cannot simply be copied, and weights sparse strokes and the predicted clean endpoint.

Write-read cycle. The one-step endpoint $\hat{x}_0 = y_\tau - \tau\hat{u}_\omega$ is reread through the frozen target retina. A visual NCE cycle objective aligns it to independent target renderings:

$$\mathcal{L}_{\text{cycle}} = \text{NCE}\left(R_{\bar{\theta}}(\hat{x}_0), R_{\bar{\theta}}(x_{t+1}^{(1)}), R_{\bar{\theta}}(x_{t+1}^{(2)})\right).$$

The complete loss combines flow, energy, cross-render invariance, retina contrast, endpoint, stroke, and cycle terms. At inference, the writer samples a candidate set $\mathcal{C}_t$ and selects

$$\hat{x}_{t+1} = \arg \max_{c \in \mathcal{C}_t} E_\psi(h_t, z_t, R_{\bar{\theta}}(c)).$$

The selected pixels become the next observation. This closes the visual loop and turns autonomous stability into a mandatory model property rather than a post-processing concern.

Model-induced visual rollout. V6 invokes the same sampler and energy reranker during training. From a clean prefix it generates two short-flow candidates, selects and detaches one bitmap, rereads it with the online retina, and advances the recurrent state. For two rollout steps it minimizes

$$\mathcal{L}_{\text{roll}} = 0.15\mathcal{L}_{\text{state}} + 0.35\mathcal{L}_{\text{energy}}^{\text{roll}} + 0.30\mathcal{L}_{\text{recovery}}.$$

The terms align clean and induced recurrent states, predict the real next image from the induced state, and recover clean next pixels after generated feedback. All targets remain continuous image fields. Candidate selection is stop-gradient, which bounds memory while exposing the trainable state and writer to deployed pixels.

Calibrated context and sampled identity. V7 corrects two training/evaluation mismatches. Raw target-energy differences can be increased by changing the scale or all competing scores, so V7 uses the normalized visual target log probability

$$\ell(y \mid h, \mathcal{C}) = s(y \mid h) - \log \sum_{c \in \mathcal{C}} \exp s(c \mid h)$$

and minimizes

$$\mathcal{L}_{\text{ctx}} = \max\{0, m - \ell(y \mid h_{\text{full}}, \mathcal{C}) + \ell(y \mid h_{\text{last}}, \mathcal{C})\}.$$

Predictive Visual Field

Language evolves as a distribution over continuous retinal states; a separate visual actuator writes each state as ink.

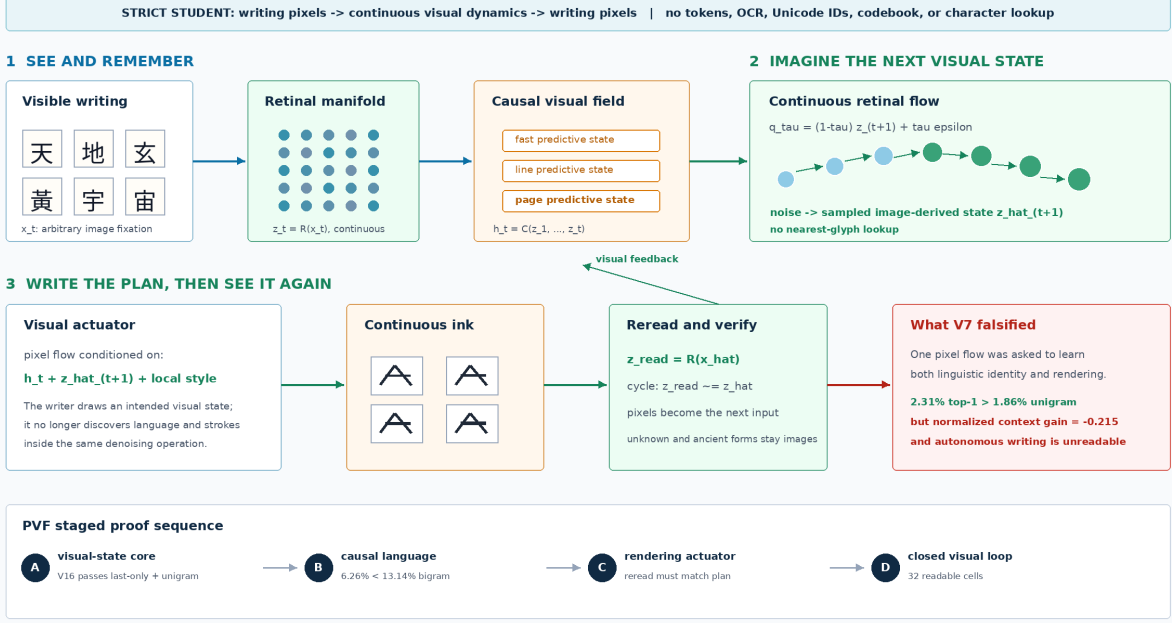


Figure 3: Predictive Visual Field factorization. A causal image field predicts a low-variance continuous visual proposal and a stochastic distribution over image-derived retinal states. A later separate actuator renders a selected state and the retina rereads its pixels.

The last-only branch is detached. The training-only set $\mathcal{C}$ contains independent image renderings; positives are inferred by retinal similarity. Offline strings select the curriculum but anchor IDs and target indices do not enter the student, the pixels are disjoint from evaluation views, and the anchors are not deployed. V7 also differentiates through a two-step numerical pixel-flow endpoint,

$$\mathcal{L}_{\text{sample}} = \text{NCE} \left(R_{\hat{\theta}}(\hat{x}_0^{K=2}), \{R_{\hat{\theta}}(y^{(v)})\}_{v=1}^V \right).$$

This trains the sampled image path rather than only a one-step endpoint proxy.

4 Predictive Visual Field

Figure 3 separates linguistic uncertainty from stroke rendering. For target state $z_{t+1} = R(x_{t+1}) \in \mathbb{S}^{d-1}$, a deterministic proposal predicts

$$\mu_t = \frac{P_{\psi}([h_t, z_t])}{\|P_{\psi}([h_t, z_t])\|_2}.$$

The proposal is a continuous image-derived state, not a class index. It is trained with geodesic alignment, image contrast, and full-history advantage against a detached last-image branch.

In parallel, choose random unit source ϵ , let q_{τ} follow the shortest spherical geodesic from z_{t+1} to ϵ , and let u_{τ} be its tangent velocity. The conditional field learns

$$\mathcal{L}_{\text{state-flow}} = \mathbb{E} \|v_{\eta}(q_{\tau}, \tau, [h_t, z_t]) - u_{\tau}\|_2^2.$$

The loss is full tangent-vector energy, not coordinate-mean MSE. Sampling integrates backward with the spherical exponential map,

$$q_{k-1} = \text{Exp}_{q_k}(-\Delta\tau v_\eta(q_k, \tau_k, [h_t, z_t])),$$

yielding intended next visual states $\hat{z}_{t+1}$, not glyph IDs. Candidate images are used only by the evaluator: it rereads them with the frozen retina and computes kernel density against continuous samples. There is no learned 512-way output layer.

The training-only multiview anchor curriculum contains image tensors. Positives are inferred from retinal cosine similarity; labels and target indices do not enter the student, and the bank is not deployed. This visual contrast is necessary because geodesic endpoint alignment alone can reward a visual mean without separating language alternatives.

V16 augments the pretrained recurrent trajectory $h_{1:t}^R$ with three residual causal memory blocks. Each block combines a left-padded depthwise temporal field at dilation 1, 2, or 4, global causal attention, and a gated feed-forward residual. For memory M_ϕ , the corrected state is

$$h_t^{V16} = h_t^R + \sigma(g)W_o \text{LN}\{M_\phi(h_{1:t}^R, z_{1:t})\}.$$

The per-channel gate is initialized at $g = -4$, preserving the trained V15 function at the start of the upgrade. The selected checkpoint has mean $\sigma(g) = 0.018023$; therefore the frozen result is a bounded memory pilot, not evidence that a large residual correction is necessary.

V17 implements a separate visual actuator that writes

$$\hat{x}_{t+1} = W_\omega(\epsilon_x, h_t, \hat{z}_{t+1}, R_\theta(x_t))$$

and is trained with pixel flow plus a reread constraint

$$\mathcal{L}_{\text{act}} = \mathcal{L}_{\text{pixel-flow}} + \lambda[1 - \cos\{R_{\hat{\theta}}(\hat{x}_{t+1}), z_{t+1}\}].$$

No nearest-glyph projection, character classifier, or token unembedding is permitted. In V17, $\hat{z}_{t+1}$ is supplied by a different-font image of the target form so actuation is isolated from language prediction. This differs from Embedded Language Flows, which demonstrates effective continuous flow dynamics but encodes and finally unembeds discrete tokens [37]. It combines the next-fixation representation objective of recurrent JEPA [35] with the prediction/generation separation explored by D-JEPA [36], and uses intrinsic flow matching on a Riemannian manifold [32, 33], while retaining an image-derived language state.

5 Isolated Visual Actuation and Motor Planning

The V17 actuator tests whether an image-derived continuous state can control new pixels without a glyph table. A frozen retina produces $z = R(x^{\text{semantic}}) \in \mathbb{S}^{191}$ from one font view, and a convolutional encoder produces style vector $s = E_\omega(x^{\text{style}})$ from a different character in the desired target render. A 5.73M-parameter conditioned U-Net integrates rectified flow from Gaussian noise to a 32×32 ink image. The target image enters losses only; the spatial condition to the writer is a constant blank plane.

Training combines stroke-weighted flow matching, an endpoint loss, retinal cosine, multi-positive retinal contrast, and gradients through a two-step deployed sampler. At evaluation, correct and shuffled branches receive identical style images and initial noise. Only the intended state is rolled within the batch. The global evaluator retrieves among all 512 generated examples, with duplicate positives inferred from image-state similarity rather than labels.

PVF V16: Causal Visual Memory on One RTX 4090

A 16.47M-parameter student reads writing pixels, forms continuous visual states, and predicts the next state without a symbolic vocabulary.

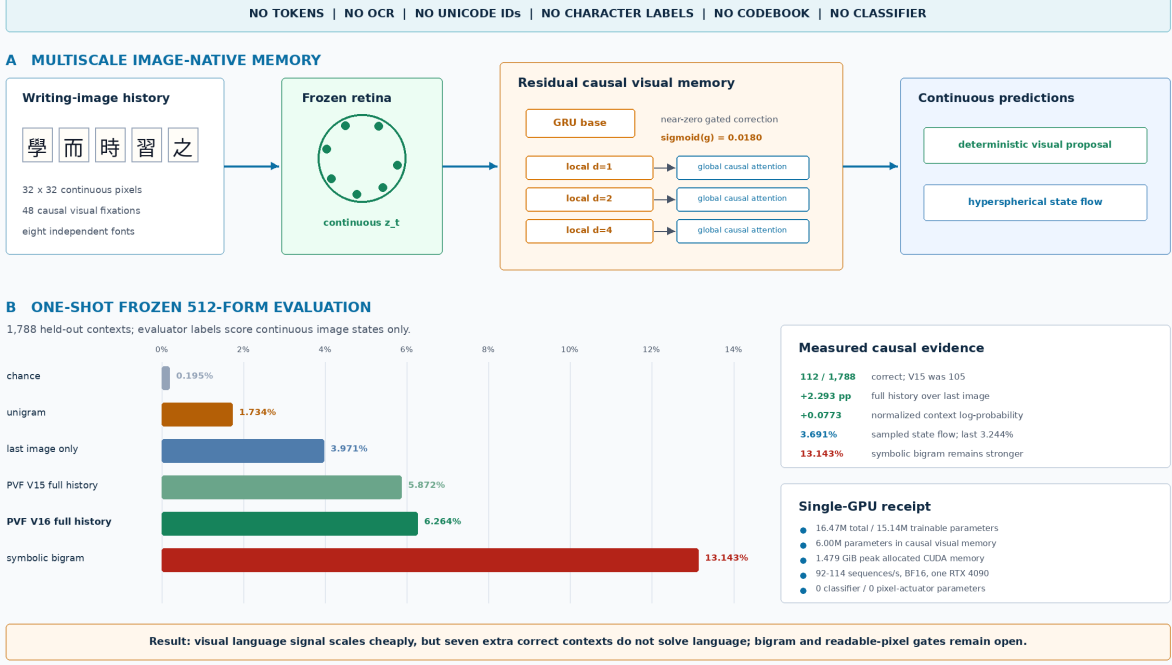


Figure 4: Measured PVF V16 memory result. Both deterministic proposal and stochastic hyperspherical field use full image history and beat random, last-only, and unigram controls. V16 improves proposal top-1 by seven frozen contexts over V15, but remains below the symbolic bigram and has no pixel actuator.

V17 was selected at step 1,600 by a preregistered development rule and queried the frozen record split once. Correct-state top-1 is 58.594%, versus 0.977% shuffled; target cosine is 0.71301 versus 0.08612; and style-copy cosine is only 0.05858. These measurements reject style copying as the explanation for the causal signal. Pixel F1 is 0.43849 versus 0.28001 shuffled, failing the required 0.50, and human review rejects most outputs as unreadable. Training takes 339.67 seconds and peaks at 1.588 GiB allocated CUDA memory on one RTX 4090.

The result identifies a topology gap rather than an identity-control gap.

5.1 Topology-first visual motor plan V18

V18 decodes state and style into a directly supervised spatial visual motor plan,

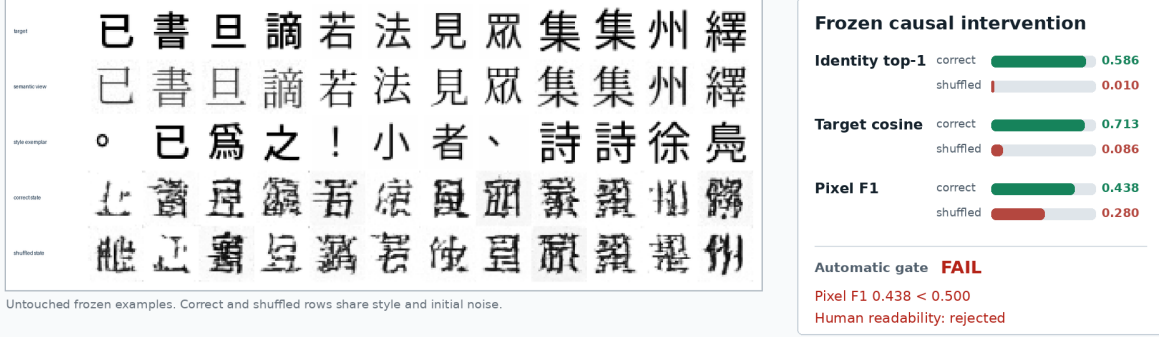
$$p_\omega = D_\omega(z, E_s(x^{\text{style}})) \in [0, 1]^{H \times W}.$$

A context MLP modulates a learned 4×4 seed, conditioned residual blocks upsample it to 32×32 , and a sigmoid emits continuous ink. The loss joins stroke-weighted binary cross entropy, soft Dice, pixel and Sobel-edge distances, retinal identity and contrast, and a correct-versus-shuffled state margin. Target pixels supervise the output but never enter the condition. The plan remains continuous and image-derived; it is not copied target ink, an ID lookup, or a discrete visual codebook. Stochastic flow is reserved for later residual style and surface variation rather than stroke placement.

V17: continuous visual state controls pixels, not yet readable strokes

One RTX 4090 | 5.73M trainable actuator parameters | no token IDs, OCR, Unicode IDs, labels, or glyph lookup

different-font image -> frozen retina -> continuous intended state + style image -> pixel flow -> generated ink -> reread



Measured verdict visual identity is causally controlled; exact stroke topology and readability are not solved

Next: decode the continuous state into a supervised spatial motor plan, then use stochastic flow only for refinement

Figure 5: Measured V17 frozen result. Continuous intended state strongly controls recovered visual identity under a same-noise shuffled-state intervention, but pixel F1 and human readability fail. The correct-state row contains state-dependent pseudo-characters, not reliably readable target forms.

The salted partition contains 6,573 training, 223 development, and 221 frozen records. Selection maximized development correct-state pixel F1 subject to fixed topology, state-dependence, retinal-similarity, and non-copying gates. Step 1,400 was selected at pixel F1 0.6890 versus 0.3368 shuffled and global identity top-1 78.32% versus 1.17%. A fresh development seed yields 512 examples: identity top-1 is 73.63% versus 0.98%, target cosine is 0.8462 versus 0.0716, and pixel F1 is 0.6577 versus 0.3129. Mean ink fraction is 0.17505 in both branches and style-copy cosine is 0.0686.

Full-resolution review finds clear simple and medium forms but merged strokes in dense forms. Because the preregistration required human readability without defining a numeric blinded threshold, post-hoc inspection cannot authorize a frozen query. V18 therefore accepts readable topology emergence on development, rejects broad-complexity readability, and leaves the frozen identifier bank sealed. The next experiment must define complexity strata and a recognition rubric prospectively.

5.2 Spatial-retinal residual V19 and its rejection

V18 compresses the semantic image into one global vector before drawing. V19 tests the narrower hypothesis that this pooling discards component location in dense forms. The frozen retina now exposes both its existing global state and its pre-pooling spatial field,

$$(z, F) = R(x^{\text{semantic}}), \quad z \in \mathbb{S}^{191}, \quad F \in \mathbb{R}^{192 \times 4 \times 4}.$$

V18: a 2.36M visual motor plan learns readable writing topology

One RTX 4090 | continuous image-derived intent | no tokens, OCR, Unicode IDs, labels, lookup, or codebook

different-font image -> frozen retina -> continuous intent + style image -> deterministic spatial motor plan -> ink



Fresh development renderings. Correct and shuffled rows share style; only continuous intent changes.

Measured consequence

structured writing is learnable as continuous image topology with a small consumer-GPU model
autonomous language choice, complex-form fidelity, and closed-loop readability remain unsolved

DEVELOPMENT-ONLY EVIDENCE | selected step 1,400 | 1,600 updates | 458.34 s | 0.778 GiB peak allocated CUDA

Figure 6: Measured V18 development result. A 2.36M-parameter deterministic motor planner turns continuous image-derived intent into recognizable writing. Shuffling only intended state destroys identity and topology while preserving mean ink occupancy. This is development-only evidence; the frozen bank remains untouched.

Let $c = [z, E_s(x^{\text{style}})]$ and let $q_0 = S(c)$ be the learned global 4×4 seed of a V18-architecture planner. A conditioned convolutional adapter supplies a spatial residual,

$$a = A_\varphi(F, c), \quad q = q_0 + \sigma(g)a, \quad p = D(q, c).$$

The adapter’s final projection is initialized exactly to zero, so V19 initially equals the selected global planner for every input even though its scalar gate is trainable. All global-planner and retina parameters remain frozen. This makes a gain attributable to spatial information rather than unrestricted baseline fine-tuning.

A clean causal comparison requires a new global baseline trained from scratch on the V19 partition. Loading V18 would contaminate V19 because V18’s training split can intersect the new holdout. The selected clean global checkpoint is therefore frozen before the spatial adapter is trained; no V17 or V18 learned weight enters the experiment.

Complexity is computed from target image geometry, never a character label. For thresholded ink b , define

$$C(b) = \bar{b} + \frac{1}{2} \left(|\overline{\Delta_x b}| + |\overline{\Delta_y b}| \right) + 0.1 \mathbf{1}[\text{AvgPool}_4(b) > 0.05].$$

The preregistered strata are simple $C < 0.24$, medium $0.24 \leq C < 0.35$, and dense $C \geq 0.35$, weighted 1, 1.25, and 2 during spatial training. The objective is

$$\begin{aligned} \mathcal{L}_{19} = & \mathcal{L}_{\text{WBCE}} + \mathcal{L}_{\text{Dice}} + 0.5\mathcal{L}_1 + 0.25\mathcal{L}_{\text{edge}} \\ & + 0.05\mathcal{L}_{\text{id}} + 0.05\mathcal{L}_{\text{NCE}} + 0.1\mathcal{L}_{\text{spatial-margin}} + 0.1\mathcal{L}_{\text{zero-margin}}. \end{aligned}$$

V19 development gate	Measured	Required	Result
Overall pixel F1	0.6710	> 0.68	fail
Dense pixel F1	0.7278	> 0.58	pass
Dense gain over shuffled field	0.0088	> 0.12	fail
Dense gain over zero field	0.0054	> 0.03	fail
Identity top-1	72.66%	> 75%	fail
Target cosine	0.8416	> 0.84	pass

Table 1: Fresh 512-candidate V19 development audit. Four fixed automatic gates fail, so the blinded human review and frozen evaluation are not authorized.

Rowwise terms use the complexity weights. The margins require correct fields to outperform batch-shuffled fields by 0.03 and the zero-field branch by 0.01 in target pixel error.

Evaluation fixes target and style while comparing five branches: correct global state plus correct field; correct global plus shuffled field; shuffled global plus correct field; both shuffled; and correct global plus zero field. The prospective gate requires overall pixel F1 above 0.68, dense F1 above 0.58, dense spatial-shuffle and zero-field advantages, global identity above 0.75, and 38/48 blinded recognitions with stratum-specific minima. The implementation passes exact-equivalence, gradient-isolation, field-shape, intervention, complexity, and serialization tests.

A clean global planner was trained from scratch for 1,600 updates on the V19 partition, selected at step 1,600, and frozen. It reached development F1 0.6754, identity top-1 77.93%, and target cosine 0.8339. The V19 intervention then trained only 764,545 adapter parameters for 1,600 updates in 329.51 seconds, using 0.899 GiB peak allocated CUDA memory on one RTX 4090. A separate deterministic audit rendered 512 new development candidates. Table 1 records the fixed decision.

This result rejects the additive routing mechanism, not the possibility that the field contains useful information. The global planner can already draw the whole form, so ordinary reconstruction gradients need not use the field. The next bounded candidate must make local visual state structurally necessary.

5.3 Retinal topology router V20: local causality without writer selection

V20 implements the preregistered coarse/detail decomposition

$$p = \sigma(U(G(z, s)) + (I - UD)H(F, s)),$$

where G emits only one logit per 8×8 output block and H maps each 4×4 retinal-field cell pointwise to its corresponding 8×8 detail patch. The fixed projection $I - UD$ removes each patch mean, so the global route cannot represent within-block detail. An exactly capacity-matched control replaces F with $\text{tile}(z)$ while retaining the same local decoder, losses, and operations. Each arm has 506,448 trainable parameters.

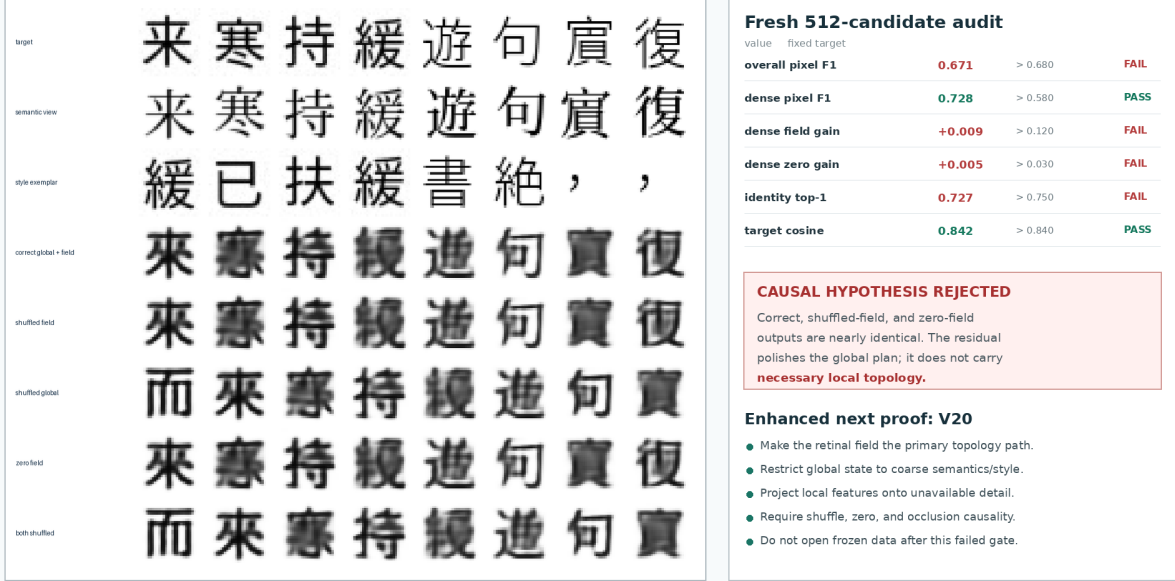
The new salted split contains 6,611 training, 196 development, and 210 frozen records. Both arms train for 1,600 updates against the same frozen V16 retina. The fixed interventions hold target and style constant while shuffling the local field, zeroing it, shuffling global state, shuffling both, or occluding each field quadrant. Validation renders 512 development candidates, including 146 dense forms. Table 2 records the final endpoints; neither is a selected checkpoint.

The candidate’s correct dense F1 exceeds shuffled-field by 0.12177 and zero-field by 0.36128; its quadrant locality is exactly 1.0. These measurements establish that the continuous field carries necessary, topographically aligned detail. They do not establish a selected writer. Overall F1 and reread cosine remain below threshold. Both arms also exceed the strict 10^{-6} detail-mean tolerance

V19: spatial retinal residual fails its causal topology test

One RTX 4090 | 764,545 trainable parameters | fixed protocol | frozen split sealed

global visual plan + gated 4x4 retinal-field residual -> ink | correct / shuffled / zero-field interventions



DEVELOPMENT-ONLY NEGATIVE RESULT | 1,600 updates | 329.51 s | 0.899 GiB peak CUDA | human audit not authorized

Measured consequence: adding spatial features is insufficient when a complete global writer can ignore them.

Figure 7: Measured V19 result. Correct, shuffled-field, and zero-field rows are nearly identical, while shuffling the global state changes identity. The spatial adapter learned a shared residual polish rather than necessary local topology.

after detail magnitude grows, indicating that floating-point recentering is a poor way to encode an algebraic invariant. Candidate dense F1 exceeds the control by only 0.01791 at the final endpoint, below the paired 0.03 margin. Candidate and control train in 325.90 and 327.11 seconds and peak at 0.323 and 0.397 GiB allocated CUDA memory, respectively.

5.4 Field-complete writer V21: complete local route, rejected rasterizer

V21 tests the field-complete correction rather than widening V20. Global state and image-derived style emit only spatially uniform modulation. Each local field cell independently emits one coarse coefficient and 63 detail coefficients,

$$(\gamma, \beta) = M(z, s), \quad (c_{ij}, a_{ij}) = A_F(\gamma \odot F_{ij} + \beta), \quad \ell_{ij} = c_{ij}\mathbf{1} + B_0 a_{ij},$$

where B_0 contains the 63 non-DC columns of a normalized Walsh–Hadamard basis. Its column sums and Gram error are exactly zero in the recorded FP32 computation. The 16 nonoverlapping 8×8 patches make the 32×32 logit field. There is no coordinate input, position parameter, cell mixing, global spatial projection, or learned glyph table. The matched control tiles z into all 16 source cells while retaining every parameter and operation. Each arm has 582,336 trainable parameters.

The preregistered salted split has 6,613 training, 195 development, and 209 frozen records. Both arms train for exactly 1,600 updates. No candidate checkpoint passes every selection gate. Table 3

V20 development gate	Candidate	Required	Result
Overall pixel F1	0.6361	> 0.66	fail
Dense pixel F1	0.7013	> 0.68	pass
Dense gain over shuffled field	0.1218	> 0.12	pass
Dense gain over zero field	0.3613	> 0.10	pass
Identity top-1	75.00%	> 70%	pass
Target cosine	0.8152	> 0.82	fail
Occlusion locality	1.0000	> 0.40	pass
Detail block-mean magnitude	2.03×10^{-6}	$< 10^{-6}$	fail
Dense gain over matched control	0.0179	> 0.03	fail

Table 2: Final V20 development endpoints. The field candidate passes every local-causality gate but fails writer selection and the matched-control margin. Because neither arm selects, the formal paired audit, blinded review, and frozen evaluation remain forbidden.

V21 candidate development gate	Measured	Required	Result
Overall pixel F1	0.6038	> 0.66	fail
Simple pixel F1	0.5648	> 0.58	fail
Medium pixel F1	0.5945	> 0.60	fail
Dense pixel F1	0.7053	> 0.70	pass
Dense gain over shuffled field	0.1739	> 0.15	pass
Dense gain over zero field	0.3465	> 0.20	pass
Identity top-1	79.10%	> 74%	pass
Target cosine	0.8331	> 0.82	pass
Occlusion pixel change	0.0926	> 0.03	pass
Occlusion locality	1.0000	> 0.95	pass
Detail block-mean magnitude	3.87×10^{-7}	$< 5 \times 10^{-6}$	pass
Decomposition error	0.0	$< 10^{-6}$	pass

Table 3: Best V21 candidate diagnostic snapshot. It passes every causal, identity, reread, and algebraic gate but fails overall, simple, and medium pixel fidelity. It is not a selected checkpoint.

therefore reports step 1,400 only as the best diagnostic snapshot under the fixed ranking, not as a selected model.

The tiled-global control selects its structural step 1,200 and reaches only 0.1473 overall F1, 0.3074 dense F1, 25.00% identity top-1, and 0.4332 target cosine. Candidate step 1,400 exceeds those values by 0.4564 overall F1 and 0.3979 dense F1. This comparison is descriptive rather than the formal paired audit: the snapshots use different development draws and steps, and the paired evaluator requires a selected candidate before constructing a shared audit. It must therefore refuse V21.

Candidate and control train in 327.04 and 333.41 seconds at 0.325 and 0.400 GiB peak allocated CUDA memory. Frozen-image count remains zero. V21 therefore establishes complete, necessary, topographic use of continuous local state, but rejects nonoverlapping pointwise patch emission as a broadly readable writer. The next writer test should add overlapping local support with a fixed partition-of-unity blend, or an equivalently bounded multiscale local operator, while retaining an exact-parameter tiled-global control and a measured influence radius.

V20: local retinal topology becomes necessary, writer gate still fails

One RTX 4090 | 506,448 parameters per arm | fixed development protocol | frozen split sealed

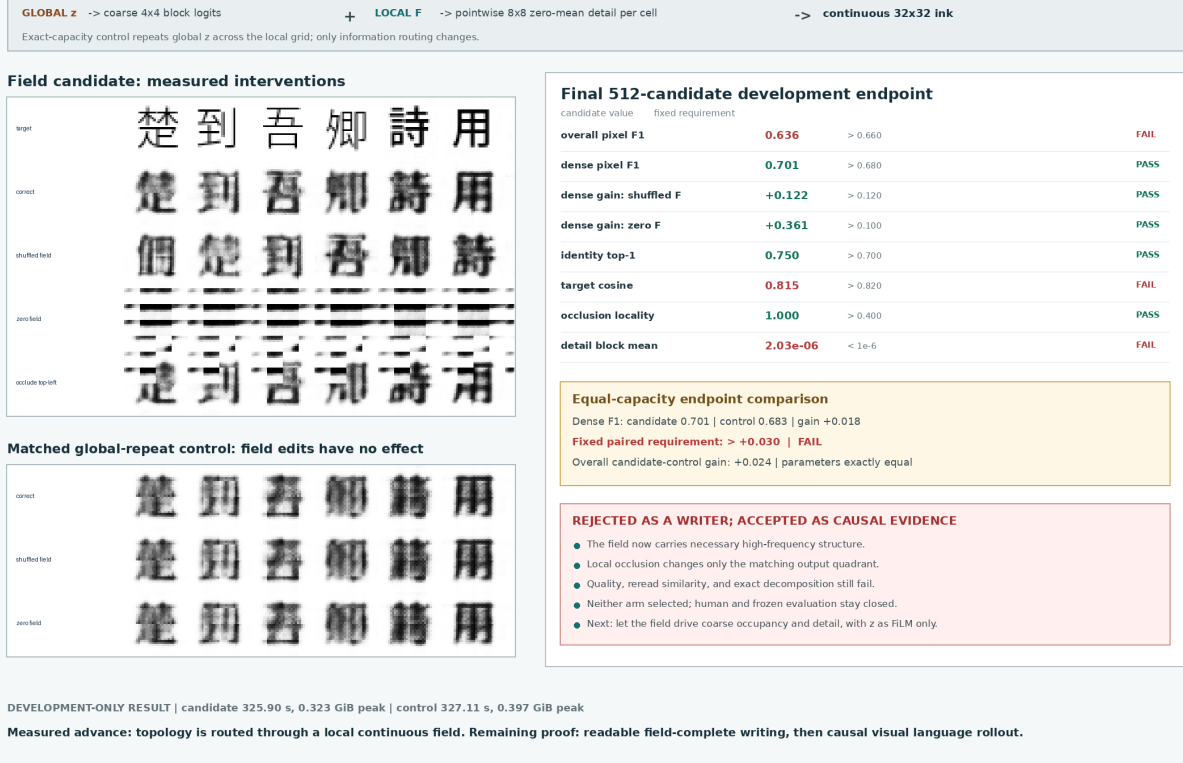


Figure 8: Measured V20 result. Shuffling or zeroing the candidate’s local field destroys detail, and a quadrant occlusion changes only its corresponding output quadrant. The equal-parameter global-repeat control is invariant to field interventions. Recognizable endpoints do not pass the fixed writer and paired-control gates.

5.5 Visual Binding Stream V22: image prompt, rejected relation

V22 removes supplied semantic intent and asks the smallest prompt-following question that also requires arbitrary visible form. Every episode is a six-frame image stream,

$$X = (l_1, g_1, l_2, g_2, o, q),$$

where two visible labels bind two distinct, previously unseen glyph images. The operation image is either the visible form for same or other, and the final query is one of the two label images. A paired counterfactual changes only q ; the correct canonical answer image must switch. Source glyphs use independent noncanonical faces, so exact pixel copying cannot satisfy the target.

The frozen V16 retina maps each frame to global and local continuous states. A four-block visual Transformer forms a query from its final hidden state and a softmax over all six frames. Weighted global/local values feed an overlapping 12×12 , stride-8 local writer with a fixed positive partition-of-unity blend. The query-blind control replaces only the final query retinal state with a learned continuous null state. Both arms contain exactly 3,410,128 trainable parameters with identical shapes. The student path contains no string, token or Unicode ID, OCR, character/operation label, answer index, glyph lookup, finite codebook, or external language model.

The fixed identity bank contains 815 training, 104 development, and 105 frozen Chinese forms

V21: a complete local field writes Chinese structure, broad writer gate still fails

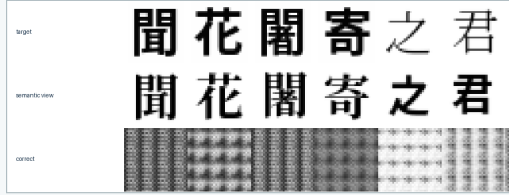
One RTX 4090 | 582,336 parameters per arm | image-only student | frozen split sealed

LOCAL F[l,j] -> shared pointwise writer -> coarse scalar + 63 zero-DC basis coefficients -> complete 8x8 patch
Global z + style provide one spatially uniform modulation. No coordinate input or cell mixing can draw around the local field.

Field-complete candidate: best diagnostic step 1400



Tiled-global control: selected structural step 1200



DEVELOPMENT ONLY | candidate 327.04 s / 0.325 GiB | control 333.41 s / 0.400 GiB | frozen images: 0

Best 512-candidate development snapshot

candidate value	fixed requirement	
overall pixel F1	0.604	> 0.660 FAIL
simple pixel F1	0.565	> 0.580 FAIL
medium pixel F1	0.594	> 0.600 FAIL
dense pixel F1	0.705	> 0.700 PASS
dense gain: shuffled F	+0.174	> 0.150 PASS
dense gain: zero F	+0.347	> 0.200 PASS
identity top-1	0.791	> 0.740 PASS
target cosine	0.833	> 0.820 PASS
occlusion locality	1.000	> 0.950 PASS
detail block mean	3.87e-07	< 5e-6 PASS

Descriptive equal-capacity comparison

Overall F1: 0.604 vs 0.147 (+0.456)
Dense F1: 0.705 vs 0.307 (+0.398)

Not a formal paired audit: the candidate did not select.

WRITER REJECTED; FIELD-COMPLETE CAUSAL ROUTE SUPPORTED

- Local state carries coarse occupancy and fine detail.
- The exact basis and every intervention invariant pass.
- The matched global-only source collapses to repeated patches.
- Simple and medium fidelity remain below fixed thresholds.
- No paired audit, human review, or frozen query is permitted.
- Next: improve local raster continuity, then predict visual answer streams.

Figure 9: Measured V21 result. The candidate uses local state for complete spatial structure: shuffling or zeroing the field corrupts the form and a local occlusion stays local. The equal-parameter tiled-global source repeats one patch. Disjoint candidate patches remain blurred and seamed, causing the preregistered broad-fidelity rejection.

under a salted split; two operation/label combinations are held out compositionally. Each arm trains for exactly 1,600 updates. Table 4 reports the final development endpoints. Candidate and control writer/identity trajectories nearly overlap, while prompt switching remains absent.

The mechanism audit makes the rejection more specific than failure to converge. Candidate mean attention is 1.0 on the operation frame, 2.49×10^{-12} and 5.53×10^{-12} on the two glyph frames, and 1.37×10^{-13} on the query. The entropy penalty rewards this confident shortcut. The task, however, is a relation among operation, query, labels, and glyphs; no single frame contains the answer. Candidate training takes 258.70 seconds at 0.315 GiB peak allocated CUDA memory, so model/GPU capacity is not the identified bottleneck.

No candidate checkpoint selects. The paired evaluator refuses before fresh data construction, blinded review is unauthorized, and no frozen image is instantiated. V22 establishes a real image-prompt to image-answer interface and weak unseen-form writing only; it does not establish prompt understanding.

V23 removes unrestricted single-frame selection. A shared visual comparator must match query and label images, while an operation-image reader provides a continuous gate,

$$a_i = C_\theta(R(q), R(l_i)), \quad m_i = \text{softmax}_{i \in \{1,2\}}(a_i), \quad s = \sigma(U_\theta(R(o))),$$

V22 candidate development gate	Measured	Required	Result
Binary choice	0.4824	> 0.85	fail
Counterfactual switch	0.0078	> 0.80	fail
Held-out-combination switch	0.0113	> 0.75	fail
Identity top-1	0.1592	> 0.45	fail
Identity gain over query shuffle	0.0225	> 0.20	fail
Target cosine	0.4601	> 0.78	fail
Pixel F1	0.5107	> 0.58	fail
Oracle-writer F1	0.6020	> 0.64	fail
Paired output L1	0.0089	> 0.08	fail
Frozen images instantiated	0	0	pass

Table 4: V22 query-aware endpoint. The anti-copy margins and structural receipts pass, but every prompt-dependence and output-quality gate fails. The query-blind control ends at 0 switch accuracy, 0.1533 identity top-1, 0.5125 pixel F1, and exactly zero paired-output difference.

$$w_i = sm_i + (1 - s)(1 - m_i), \quad (\hat{z}, \hat{F}) = \sum_{i=1}^2 w_i V(R(g_i)).$$

This relation circuit composes multiple visible roles before writing. Its operation semantics and equality function are learned from answer images; no symbolic target index enters the student. Exact query-blind, operation-blind, pair-swap, query-counterfactual, and operation-counterfactual interventions are required before any longer answer stream. A qualified writer should be frozen first so binder/writer co-adaptation cannot hide another average-glyph route.

5.6 Visual Relation Circuit V23: bounded prompt relation accepted

V23 fixes the V22 causal defect while preserving the same strict visual interface. Its bank contains 817 training, 109 development, and 98 frozen Chinese identities under the preregistered salt `visual-relation-circuit-v23`. The two held-out operation/label compositions never occur during training, although every individual operation and label pair does. Every model call receives only floating image tensors or continuous states derived from those tensors; renderer strings and episode metadata are deleted before `forward`.

Stage A independently trains a 1,122,081-parameter convolutional U-Net to map a noncanonical source-glyph image to a canonical target image. It has no identity embedding, target-font ID, coordinate channel, or character table. The selected step-1,000 checkpoint reaches 0.7583 pixel F1, 0.9951 identity top-1, and 0.9375 target cosine on all 109 unseen development identities. Its pixel-F1 gains over the raw source and a shuffled source are 0.2173 and 0.4237. The canonicalizer is then frozen before relation training.

Stage B uses normalized frozen-retina states. For each label image,

$$c_i = \langle R(q), R(l_i) \rangle, \quad m_i = \text{softmax}_{i \in \{1,2\}}(\tau c_i),$$

where the positive temperature τ is learned. A small operation-image reader produces

$$s = \sigma(U_\theta(R(o))),$$

and composes match and operation before generation:

$$w_i = sm_i + (1 - s)(1 - m_i), \quad x_r = w_1 g_1 + w_2 g_2, \quad \hat{y} = C_\psi(x_r).$$

V22: visible prompt binding fails through an operation-frame shortcut

One RTX 4090 | 3,410,128 trainable parameters per arm | image-only input/output | frozen identities sealed

Six-frame visual prompt and measured candidate selector behavior

frame 1 label 1 mean 3.4e-13 argmax 0/1024	frame 2 glyph 1 mean 2.5e-12 argmax 0/1024	frame 3 label 2 mean 1.6e-12 argmax 0/1024	frame 4 glyph 2 mean 5.5e-12 argmax 0/1024	frame 5 operation mean 100.0% argmax 1024/1024	frame 6 query label mean 1.4e-13 argmax 0/1024
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Actual endpoint samples

Only the final query image changes inside each pair; targets switch, predictions barely do.

query-aware candidate

seen	pred	target	cf pred	cf target	heldout
甲	田	乙	論	同	甲
地	歌	天	吾	同	地

query-blind control

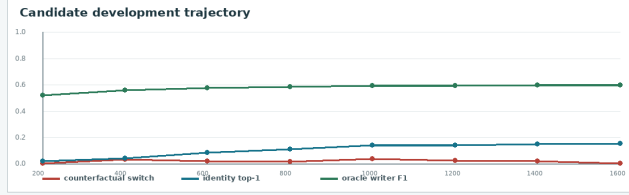
seen	pred	target	cf pred	cf target	heldout
甲	田	乙	論	同	甲
地	歌	天	吾	同	地

Fixed candidate gates at step 1,600

measured candidate	fixed requirement		
binary choice	0.482	> 0.850	FAIL
counterfactual switch	0.008	> 0.800	FAIL
held-out switch	0.011	> 0.750	FAIL
identity top-1	0.159	> 0.450	FAIL
identity gain: query	+0.022	> 0.200	FAIL
pixel F1	0.511	> 0.580	FAIL
oracle-writer F1	0.602	> 0.640	FAIL
paired output L1	0.0089	> 0.080	FAIL
target vs op. margin	+0.434	> 0.150	PASS
frozen images	0	= 0	PASS

Candidate vs query-blind control endpoint

switch 0.0078 vs 0.0000 | identity 0.1592 vs 0.1533 | F1 0.5107 vs 0.5125



DEVELOPMENT ONLY | candidate 258.70 s / 0.315 GiB | control 255.11 s / 0.316 GiB | no selected candidate

BINDING MECHANISM REJECTED

- The writer acquires weak unseen-glyph structure.
- The candidate does not bind query to either glyph.
- Entropy minimization makes the fixed operation marker
- the easiest single-frame shortcut: 1,024/1,024 argmaxes.
- No candidate checkpoint selects; paired audit refuses.
- No human review or frozen identity image is permitted.
- Next: explicit multi-frame visual relation composition.

Figure 10: Measured V22 result. Actual endpoint samples show that targets switch when only the final visual query changes, while candidate predictions barely move and match the query-blind control. The candidate selector assigns every one of 1,024 development argmaxes to the operation frame. Writer metrics improve without relational prompt binding.

Thus same routes the visually matched source and other routes its complement. The operation semantics are learned from answer-image loss, not supplied as a class label. The routed variable is a continuous source-pixel field, and C_ψ is the frozen Stage A canonicalizer. The trainable relation circuit has 25,602 parameters: the operation reader, one temperature scalar, and two continuous null states.

Candidate, query-blind, and operation-blind arms have exactly identical parameter names and shapes. Query-blind replaces only $R(q)$ with its learned continuous null state; operation-blind replaces only $R(o)$. All arms process base, query-counterfactual, operation-counterfactual, and pair-swapped image prompts jointly for 600 fixed updates. The selected candidate reaches 0.9946 binary choice, 0.9902 query switch, 0.9902 operation switch, 0.9873 identity top-1, and 0.7548 pixel F1 on checkpoint-selection development data.

A fresh paired audit then constructs 1,024 new development episodes and 4,096 prompt variants. Table 5 reports the exact interventions. The candidate changes identity and pixels when either visible causal factor changes. Each blind control is exactly invariant to the factor it cannot see, and all arms remain exactly invariant to pair order.

After the paired gate passes, an opaque development review presents 48 image cards without transcriptions, targets, correctness labels, or frozen examples. The reviewing agent chooses which

Fresh paired metric	Relation-aware	Query-blind	Operation-blind
Binary choice	0.9990	0.5010	0.5002
Query switch	0.9980	0.0000	0.0615
Operation switch	0.9980	0.0156	0.0000
Identity top-1	0.9951	0.2947	0.2993
Pixel F1	0.7563	0.5407	0.5404
Query-output L1	0.1915	0.0000	0.0109
Operation-output L1	0.1912	0.0017	0.0000
Pair-swap output L1	0.0000	0.0000	0.0000

Table 5: V23 fresh paired development audit. A control has exactly zero response to its blinded factor by construction, while the relation-aware arm retains both dependencies with matched parameter shapes.

V23 frozen gate	Measured	Required	Result
Binary choice	0.9983	> 0.95	pass
Query switch	0.9961	> 0.90	pass
Operation switch	0.9971	> 0.90	pass
Held-out minimum switch	0.9961	> 0.85	pass
Identity top-1	0.9946	> 0.75	pass
Pixel F1	0.7848	> 0.68	pass
Target cosine	0.9399	> 0.82	pass
Query-label visual match	0.9995	> 0.98	pass
Operation-gate accuracy	1.0000	> 0.98	pass
Pair-swap consistency	1.0000	> 0.99	pass

Table 6: Single authorized V23 frozen evaluation over previously unseen identities. Every threshold was fixed before implementation and remained unchanged.

of the two visible source forms the generated answer depicts before the scorer opens the sealed key. It obtains 48/48 overall and 12/12 on held-out compositions, above the fixed 44/48 and 11/12 gates. This is an agent visual review, not a human-subject study.

Only then does the frozen evaluator instantiate the 98 held-out identities. It runs once on 1,024 episodes, 4,096 prompt variants, and four identity-bank views, with no model selection or threshold changes. Every gate in Table 6 passes. The run takes 4.74 seconds and peaks at 0.498 GiB allocated CUDA memory. Stage A training takes 214.10 seconds at 0.700 GiB; candidate relation training takes 117.18 seconds at 1.406 GiB, all on one RTX 4090.

V23 therefore accepts a bounded image-only relation: multiple visible roles are composed, unseen source forms are routed, and the answer is emitted as pixels. It does not establish arbitrary syntax or free-form generation. Frame roles and the two-pair same/other algebra remain architecturally fixed, and the answer is a canonicalized version of one visible source form. V24 implements the preregistered next test: remove absolute frame roles, consume a variable-length visual instruction, emit two answer-image frames, and make frame 2 causally depend on rereading the generated pixels of frame 1.

V23: an image-only relation circuit follows a visible prompt

One RTX 4090 | six raster frames in, one raster answer out | 98 unseen frozen identities | no token or Unicode path

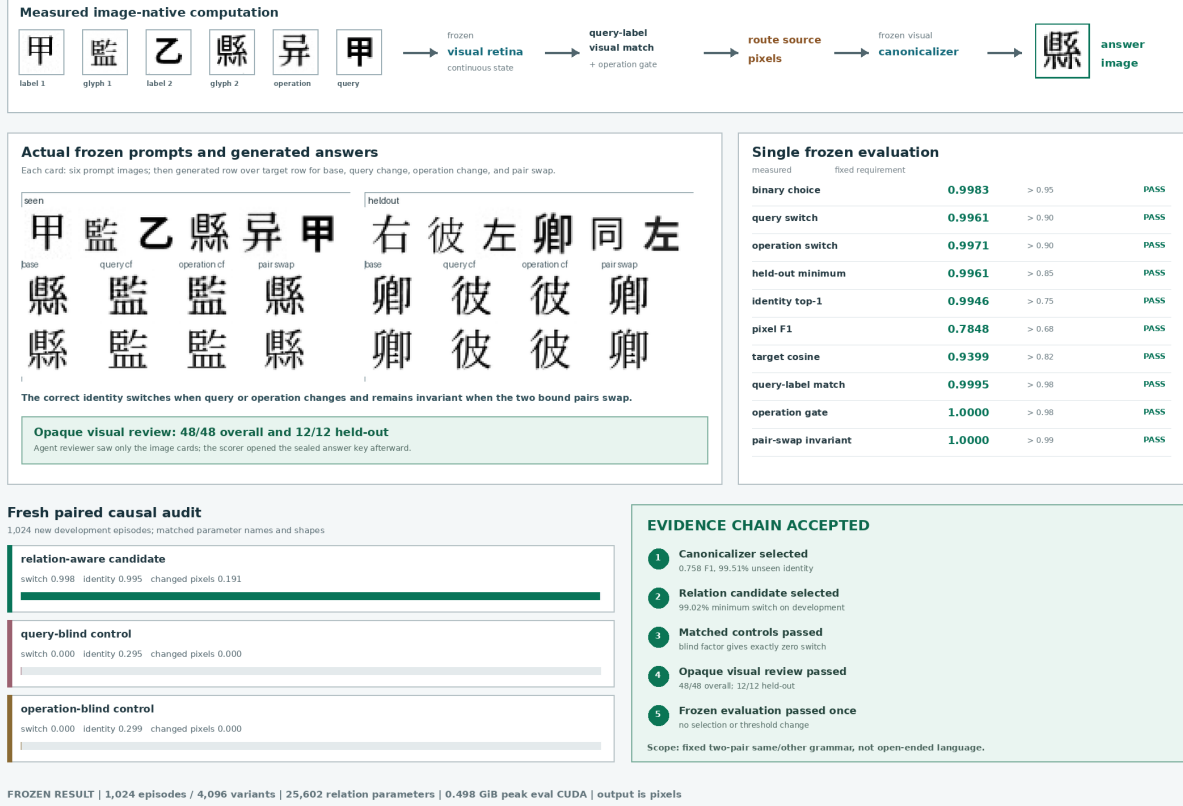


Figure 11: Measured V23 result. Actual frozen samples show generated answers above their targets for base, query-counterfactual, operation-counterfactual, and pair-swapped prompts. The matched blind controls localize both causal dependencies, and the single frozen run passes every fixed gate.

5.7 Visual Packet Reread Stream V24: causal two-frame stream accepted

V24 replaces V23’s six absolute roles with a variable sequence of three-frame visual packets. Its prompt and autonomous answer have shapes

$$X \in [0, 1]^{B \times T \times 1 \times 32 \times 32}, \quad T \in \{15, 18, 21, 24\}, \quad \hat{Y} \in [0, 1]^{B \times 2 \times 1 \times 32 \times 32}.$$

Each packet is *header*, *content A*, *content B*. The visible header glyphs for pair (U+5951), operation (U+6CD5), query (U+95EE), distractor (U+65C1), and stop (U+6B62) denote their respective roles. Packets before the terminating packet are shuffled. Training uses 15, 18, and 21 active frames; 24 frames with three distractors are held out by length. Dynamic batching adds only all-zero packet images and supplies neither active lengths nor padding masks.

The identity bank contains 829 training, 88 development, and 107 frozen Chinese forms under the fixed salt `visual-packet-reread-stream-v24`. Two operation/label-pair compositions are held out while every constituent operation and label pair appears separately in training. As in V23, renderer strings and episode metadata are deleted before every model call.

For packet j , the frozen V16 retina yields normalized header and content states H_j, A_j, B_j .

Fresh paired metric	Packet-aware	Corresponding blind
Query switch	0.9922	0.0000
Query-output L1	0.1677	0.0000
Operation switch	0.9932	0.0000
Operation-output L1	0.1676	0.0000
Frame-2 history switch	0.9961	0.0000
History-output L1	0.1421	0.0000
Minimum role localization	1.0000	0.0820
Frame-1 identity top-1	0.9961	0.3410 / 0.3197 / 0.2740
Frame-2 label top-1	0.9969	0.4873 / 0.3717

Table 7: V24 fresh paired development audit. The last two control cells list query/operation/header-blind and history/header-blind results, respectively. All arms have exactly 1,347 trainable parameters.

Three learned normalized prototypes locate roles from header pixels:

$$e_{jr} = \tau_r \langle H_j, p_r \rangle, \quad r \in \{\text{pair}, \text{operation}, \text{query}\}.$$

Straight-through hard routing selects the top two pair packets and top one operation and query packet; its backward surrogate is a softmax. Scores inspect only header images, not packet contents. With located query q , operation o , pair labels A_j , and pair-glyph images B_j^{image} , frame 1 uses the frozen V23 comparison temperature, operation reader, and canonicalizer:

$$m_j = \text{softmax}_{j \in \text{pair}} (\tau_{23} \langle q, A_j \rangle), \quad s = \sigma(U_{23}(o)),$$

$$w_j = sm_j + (1 - s)(1 - m_j), \quad x_1 = \sum_{j \in \text{pair}} w_j B_j^{\text{image}}, \quad \hat{y}_1 = C_{23}(x_1).$$

The second step rereads the actual generated pixels:

$$r_1 = R(\hat{y}_1), \quad b_j = \text{softmax}_{j \in \text{pair}} (\tau_{23} \langle r_1, B_j \rangle),$$

$$x_2 = \sum_{j \in \text{pair}} b_j A_j^{\text{image}}, \quad \hat{y}_2 = C_{23}(x_2).$$

Thus no pre-render identity state crosses the frame-1/frame-2 boundary. All weighted reductions accumulate in FP64.

V16 retina parameters and every inherited V23 parameter are frozen. V24 trains only three 192-dimensional role prototypes, four 192-dimensional continuous null states, and three role temperatures: 1,347 parameters total. The packet-aware, header-blind, query-blind, operation-blind, and history-blind arms have identical parameter names, shapes, and counts. Each receives exactly 800 AdamW updates in BF16 with batch 64 and eight workers on one RTX 4090. The candidate selects step 100, then completes the fixed run in 394.18 seconds at 2.322 GiB peak allocated CUDA memory.

A fresh audit uses 1,024 newly rendered development episodes at seed 22260832. Table 7 shows the paired causes. A blind arm is exactly invariant to the intervention it cannot observe. Header blindness prevents reliable role localization, while history blindness leaves frame 1 intact but breaks frame 2.

After every paired gate passes, an opaque agent visual audit presents 48 development cards containing only the packet images, two visible source choices, and two generated frames. It exposes

V24 frozen gate	Measured	Required	Result
Frame-1 binary choice	0.9980	> 0.95	pass
Query switch	0.9766	> 0.90	pass
Operation switch	0.9727	> 0.90	pass
Generated-history switch	0.9961	> 0.90	pass
Held-out minimum switch	0.9635	> 0.85	pass
Frame-1 identity top-1	0.9867	> 0.75	pass
Frame-2 label top-1	0.9973	> 0.95	pass
Frame-1 pixel F1	0.8371	> 0.68	pass
Frame-2 pixel F1	0.7286	> 0.58	pass
Generated-history consistency	0.9863	> 0.92	pass
Held-out $T = 24$, frame 1	0.9847	> 0.90	pass
Held-out $T = 24$, frame 2	1.0000	> 0.90	pass

Table 8: Single authorized V24 frozen evaluation. The sealed report records no model selection, no threshold change, and no repeated frozen run.

no targets, transcriptions, correctness labels, or frozen forms. The committed judgments score 47/48 for frame 1 and 48/48 for frame 2, including 12/12 for each frame at held-out $T = 24$. This is an agent visual audit, not a human-subject study.

Only then does the evaluator instantiate frozen images. It runs exactly once on 1,024 episodes, 107 unseen identities, and four identity-bank views, without model selection, threshold changes, or repetition. Table 8 reports the primary gates. All remaining role, distractor, packet-permutation, teacher-forcing, and image-boundary gates also pass. The run takes 8.13 seconds and peaks at 0.542 GiB allocated CUDA memory.

V24 therefore accepts a variable-input, two-frame image stream under one fixed visual grammar. It establishes that frame 2 can depend causally on rereading generated frame-1 pixels, not merely on a hidden state. It does not establish arbitrary sentence parsing, factual retrieval, page continuation, historical form synthesis, free-form output length, or movie generation. V25 implements the next narrower question: remove packet algebra, train on ordinary rendered Chinese, and require the resulting visual history to beat causal and frequency controls before a writer is authorized.

V24: visible packets become a causal two-frame image stream

Variable 15-24 frame prompt | generated glyph then generated label | 1,347 trainable parameters | one RTX 4090

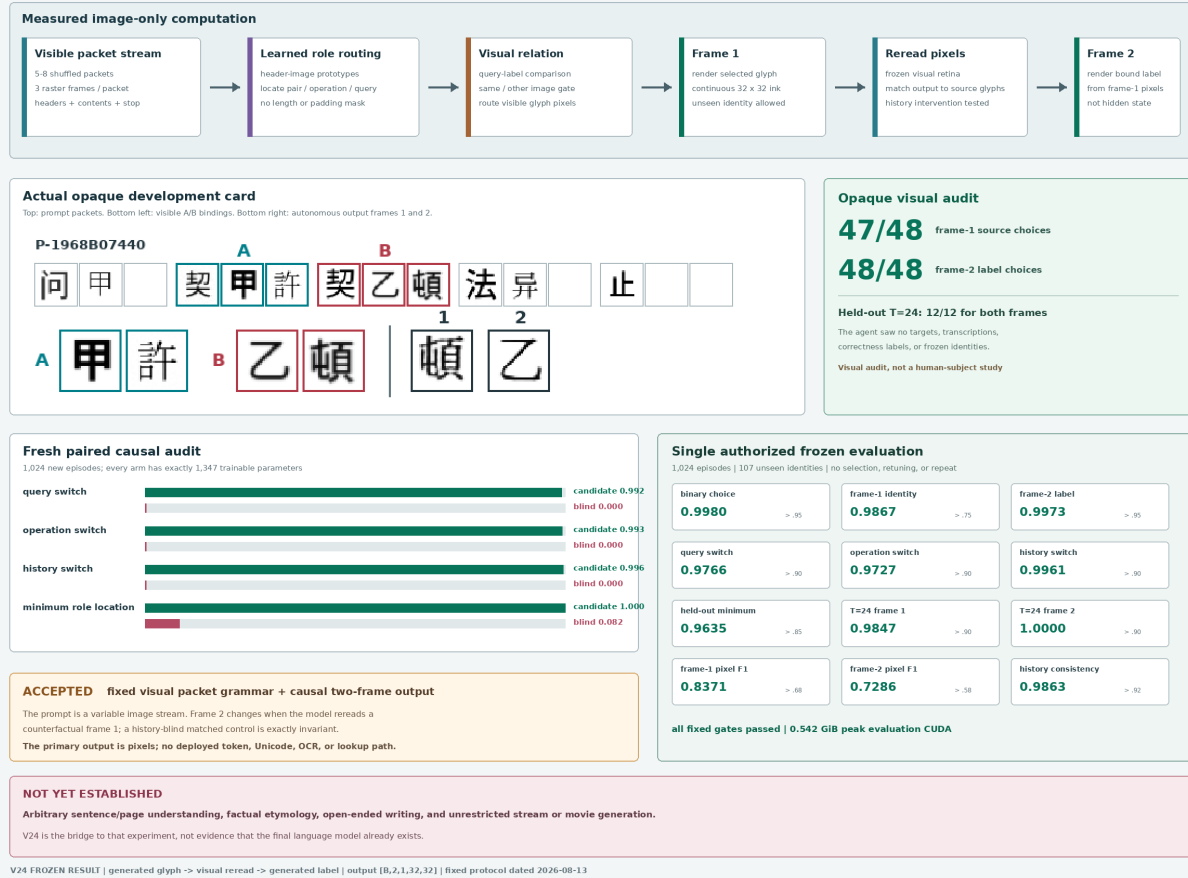


Figure 12: Measured V24 result. An actual opaque review card shows a variable-length image prompt and two autonomous answer frames. Matched controls localize header, query, operation, and generated-history causality; every gate in the single frozen evaluation passes.

5.8 Visual Cell Stream V25: natural Chinese prediction rejected

V25 treats each visible writing unit as an inspectable retinal observation and stacks observations along visual time:

$$X \in [0, 1]^{B \times T \times 1 \times 32 \times 32}, \quad T = 64.$$

The third nontrivial axis of one sample is reading order, not a character ID or geometric depth. The student receives only these grayscale images, their order, continuous corruption and flow times, random image noise, and its own generated pixels. Strings, bytes, token and Unicode IDs, OCR transcripts, character labels, candidate-bank rows, and external models are excluded from the learned path.

The fixed manifest contains 7,017 records from 16 public-domain Chinese works, with SHA-256 76048753...f6866c03. Hash partitioning by original record gives 6,608 training, 190 development, and 219 frozen groups. Training and development fonts are disjoint. Two independently rendered views of every 65-cell span reverse their context/target roles within a batch; their strings and renderer metadata are deleted before each model call. The frozen partition is counted but never rendered. An evaluator-only 1,024-form image bank and symbolic frequency tables are unavailable to the student and absent from deployment.

A frozen V16 retina maps each image to a continuous unit state $z_t \in \mathbb{S}^{191}$. Eight causal Transformer blocks of width 384 use rotary reading position and continuously corrupted prior images:

$$h_t = C_\theta(\tilde{z}_{\leq t}), \quad \hat{z}_{t+1} = \frac{P_\theta(h_t)}{\|P_\theta(h_t)\|_2}.$$

There is no character embedding or vocabulary projection. The fixed language objective combines independent-view cosine prediction with multi-positive in-batch visual contrast,

$$\mathcal{L}_{\text{language}} = 1 - \langle \hat{z}_{t+1}, z'_{t+1} \rangle + 0.5 \mathcal{L}_{\text{visual contrast}}.$$

The complete model contains 25,549,714 parameters; 15,810,241 are trainable in the language stage. The run performs 2,400 BF16 AdamW updates with physical batch three, two view directions, and four-step accumulation, for 24 effective streams per update on one RTX 4090.

Development evaluation uses 2,048 fixed windows. The same model is queried with full history, only the last cell, shuffled prior history with the last cell fixed, and blank history. The evaluator separately computes image-unigram and symbolic-bigram controls from training frequencies. It also forms 512 pairs with the same last cell but different histories and different true next cells. Table 9 reports every selection-relevant measure.

Thus the causal field uses ordered history, but does not use it accurately enough to select the correct next form. Its top-1 changes on 77.93% of paired counterfactuals, yet changes to the correct target only 12.89% of the time. Context sensitivity alone is not understanding. The language stage is rejected, writer training is not authorized by the fixed path, and no frozen evaluation is opened.

For diagnosis only, a separate command explicitly marks itself exploratory and continues from the rejected language checkpoint. It freezes the causal field and trains the 7,073,617-parameter continuous flow writer for 1,200 updates. For clean target x , noise ϵ , and time τ it learns the velocity of $y_\tau = (1 - \tau)x + \tau\epsilon$, conditions on h_t and $\hat{z}_{t+1}$, and rereads its sampled pixels. This run cannot change the evidence verdict. On 256 targets and 16 autonomous continuations, generated identity top-1 is zero, reread target cosine is 0.08015, and pixel F1 is 0.32210. Blank rate is zero and position-16 density is 0.97657 times target density. The writer therefore learns ink occupancy, not the intended continuation.

V25 development measure	Measured	Required	Result
Full-history top-1	0.01123	diagnostic	–
Last-only top-1	0.00146	diagnostic	–
Shuffled-history top-1	0.00342	diagnostic	–
Image-unigram top-1	0.01611	diagnostic	below
Symbolic-bigram top-1	0.12158	benchmark	below
Full minus last top-1	+0.00977	> 0.030	fail
Full minus unigram top-1	−0.00488	> 0.030	fail
Full minus last log probability	+0.04882	> 0.050	fail
Full minus shuffled top-1	+0.00781	> 0.015	fail
Counterfactual switch accuracy	0.12891	> 0.550	fail
Full-history target cosine	0.27513	> 0.550	fail
Student boundary / peak CUDA	clean / 0.598 GiB	clean / < 18	pass

Table 9: V25 fixed natural-Chinese development audit. Full history has a small advantage over last-only and shuffled history, but remains below the image unigram and symbolic bigram. Six semantic and causal gates fail.

V25 localizes the next problem to the predictive state rather than page geometry. Cross-font visual appearance, exact identity, and context-predictive meaning share one target state; cosine can improve toward a frequent visual centroid without selecting the next glyph. A subsequent preregistered model should separate an exact appearance field from a context-predictive residual and require both under matched state-shuffle interventions.

Long context has a separate exact representation. For a lattice width K , sequence index t maps serpentine-wise to

$$r = \lfloor t/K \rfloor, \quad c = \begin{cases} t \bmod K, & r \text{ even,} \\ K - 1 - (t \bmod K), & r \text{ odd.} \end{cases}$$

This bijection preserves adjacency at row turns and maps clean $N \times 1 \times 32 \times 32$ cells to either an inspectable flat page or a learned $C \times R \times K$ retinal lattice with a validity mask. A 256×256 lattice holds 65,536 cells, but was not used in V25. Scaling this representation before repairing the 64-cell objective would increase context without establishing better language.



Figure 13: Measured V25 development result generated directly from the fixed language report, separately labeled exploratory writer report, and actual writer sample. Ordered visual history has a weak effect, but the unigram and bigram controls remain stronger and generated images are not the intended Chinese forms. The frozen partition stays sealed.

5.9 Factorized Visual Context V26: changed state, failed binding

V26 tests the repair proposed by V25 while preserving the same ordinary-Chinese corpus, record-group partitions, disjoint font partitions, 64-cell horizon, frozen V16 retina, and strict student boundary. It changes the supervised unit. Every natural training example predicts only its full-context endpoint and future offsets 2, 4, and 8; losses are no longer averaged over all short prefixes. A fixed set of 16,384 cross-record training pairs shares the same four-character suffix but has different earlier histories and different next glyphs. Pair suffixes are rendered with identical font and augmentation draws, so their final four image tensors are exactly equal.

For $X = (x_1, \dots, x_{64})$, the appearance route receives only the last visible image and the history route receives only the preceding images:

$$a = A_\theta(R(x_{64})), \quad r = C_\theta(R(x_{1:63})), \quad s = \text{RMSNorm}(a + \sigma(G_\theta([a; r])) \odot r).$$

Zeroing r is therefore a true last-only intervention. For two suffix-matched contexts, swapping r changes only earlier-history state while preserving the last-glyph appearance route.

For horizon $h \in \{1, 2, 4, 8\}$ and independent Gaussian noise ϵ_k , $k = 1, \dots, 8$, a shared conditional head emits unit retinal particles

$$q_{h,k} = \text{normalize } P_\theta(s + e_h + N_\theta(\epsilon_k)) \in \mathbb{S}^{191}.$$

These samples are continuous predictions, not vocabulary rows or persistent glyph prototypes. For target state y_h , the empirical energy score is

$$\mathcal{E}(Q_h, y_h) = \frac{1}{8} \sum_k \|q_{h,k} - y_h\|_2 - \frac{1}{128} \sum_{k,j} \|q_{h,k} - q_{h,j}\|_2.$$

Training combines next- and multi-horizon energy scores, continuous candidate ranking against an 8,192-entry FIFO of detached image states, and a softplus pair-ranking loss with fixed margin 0.10. The queue contains image observations but no IDs and is absent from deployed inference. This design follows the calibrated continuous-prediction motivation of proper energy scores [38]; it does not assume that particle diversity itself is semantic.

The complete model contains 19,142,721 parameters, of which 16,476,865 are trainable. Its fixed run performs 8,000 BF16 AdamW updates in 1,863.27 seconds on one RTX 4090, with 0.888403 GiB peak allocated CUDA memory. The corpus manifest SHA-256 is 76048753...f6866c03, the preregistered protocol SHA-256 is dcfa5b97...b899df1, and the final checkpoint SHA-256 is 065f84e1...db6e1d. The frozen partition is counted but never rendered.

The fixed natural audit uses 2,048 development windows and the same evaluator-only 1,024-form image bank for every branch. Table 10 shows the preregistered mechanism and language comparisons.

The retina-bank oracle is exactly 1.0, so the frozen visual sensor and evaluator can distinguish every audited target. The model nevertheless assigns almost no top-1 advantage to history. Its full-history target log probability is -6.932756 , versus -6.952485 for suffix-4 and -6.936448 after shuffling cells 1–60 while preserving the suffix. Only the full-versus-last log-probability gate passes.

The primary causal audit uses 512 cross-record suffix-4 pairs. Its result is shown in Table 11.

The pair result localizes the failure. The history branch is not constant: pixel-identical suffix contexts produce substantially different residuals. Those differences do not bind to the correct next-glyph distribution. Because suffix-matched pair members have exactly the same appearance state, ordinary two-way pair ranking and the residual-swap target test are algebraically equivalent in this implementation; both are retained to preserve the preregistered intervention interface rather

V26 natural development measure	Measured	Required	Result
Full-history top-1	0.000488	diagnostic	–
Last-only / suffix-4 top-1	0.000488 / 0.000488	diagnostic	–
Shuffled-prefix top-1	0.000488	diagnostic	–
Image-unigram top-1	0.014160	control	below
Symbolic-bigram top-1	0.135254	benchmark	below
Full minus last top-1	0.000000	> 0.020	fail
Full minus unigram top-1	−0.013672	> 0.030	fail
Full minus last log probability	+0.170451	> 0.10	pass
Full minus suffix-4 log probability	+0.019729	> 0.03	fail
Full minus shuffled log probability	+0.003692	> 0.03	fail
Full target cosine / particle spread	0.160451 / 1.210599	diagnostic	–
Retina oracle / student boundary	1.0 / clean	≥ 0.99 / clean	pass
Peak allocated CUDA	0.888403 GiB	< 18 GiB	pass

Table 10: V26 fixed natural-Chinese development audit. Full history improves target log probability over last-only, but not over the same four-cell suffix or suffix-preserving history shuffle. It remains below image unigram and symbolic bigram controls.

V26 suffix-4 pair measure	Measured	Required	Result
Suffix pixel equality	1.000000	exact	pass
Mean appearance-state difference	0.000000	exact	pass
Mean history-residual difference	4.628258	diagnostic	changed
Correct two-way pair ranking	0.500000	> 0.65	fail
Swapped-residual target accuracy	0.500000	> 0.65	fail
Mean correct-versus-other margin	0.0000677	diagnostic	near zero
Top-1 output switch rate	0.212891	diagnostic	–
Both pair targets top-1 correct	0.025391	diagnostic	–

Table 11: V26 pixel-identical suffix intervention. Earlier history materially changes the continuous residual, but neither ordinary nor swapped-residual scoring prefers the associated target above chance.

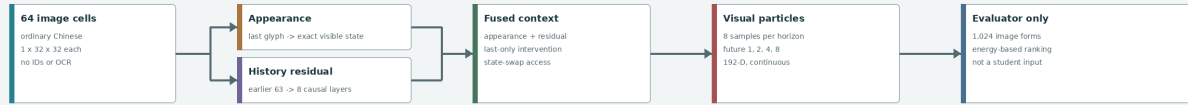
than counted as independent evidence. A suffix-8 diagnostic agrees, with 0.501953 ranking and swap accuracy.

V26 therefore rejects this factorization and objective, not image-native language modeling in general. A broad particle cloud and a changing hidden state are not evidence of useful conditional language. The next attributable experiment should remain at 64 cells and make matched target choice conditionally identifiable during optimization. It must beat suffix-only, history-shuffled, image-unigram, and symbolic-bigram controls before a pixel writer, serpentine page lattice, geometric depth, or motion is authorized.

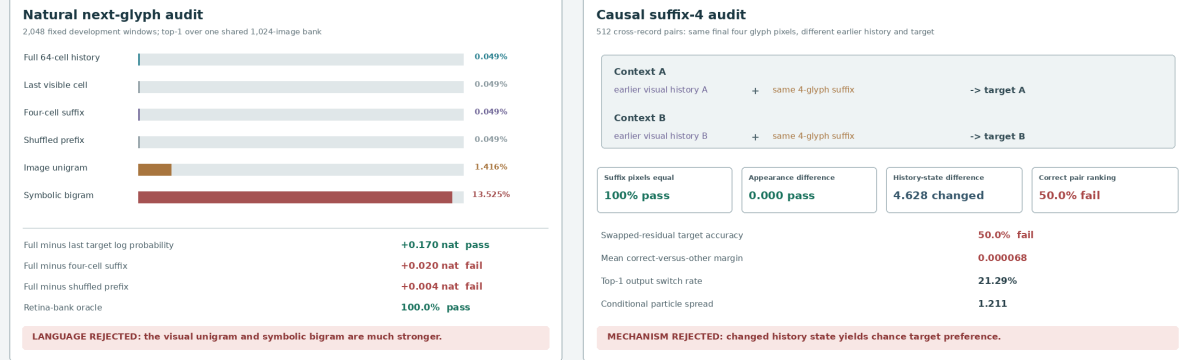
We further localize this failure with a post-hoc diagnostic that cannot revise the V26 result or authorize frozen evaluation. Every V26 parameter is frozen. Three visual scorers, totaling 1.11M trainable parameters, receive appearance, history residual, or fused state and score two arbitrary candidate images using normalized projected compatibility. One pass over all 16,384 train suffix pairs optimizes symmetric 2×2 assignment. The fixed 512 development pairs yield 1,024 cross-font assignments and 2,048 scored arms. Appearance-only accuracy is exactly 0.500000, history accuracy is 0.506836, and fused-state accuracy is 0.503418. Their mean margins are 0.000108 and 0.000371. In contrast, direct frozen-retina cosine assigns the same cross-font target images at 0.999512 with

V26: visual history changes state, but does not bind the next glyph

19.14M total parameters | 8,000 fixed BF16 updates | one RTX 4090 | 0.888 GiB peak | frozen split sealed



The appearance and history routes are separable. Suffix pairs keep the final four glyph images exactly equal while changing earlier text.



What the fixed experiment establishes

The full factorized image-only system trains in 31.1 minutes with 0.888 GiB peak allocated VRAM.
 The retina/evaluator is intact, the exact suffix intervention holds, and earlier history changes the residual state.

What V26 falsifies

Appearance/residual factorization plus energy, queue contrast, and pair ranking does not learn useful next-glyph binding.
 Internal-state sensitivity and broad visual particles are not conditional language behavior.

Next controlled question

Make target choice conditionally identifiable before adding a writer, longer context, page geometry, depth, or motion.
 A next model must beat suffix, shuffled-history, unigram, and bigram controls under paired causal intervention.

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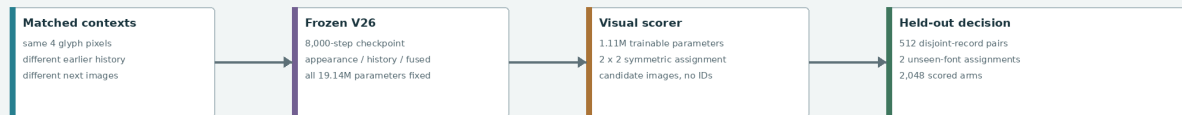
Figure 14: Measured V26 development result generated directly from the fixed audit. The visual retina and exact suffix intervention pass, and earlier history changes the residual state, but natural next-glyph accuracy remains below frequency controls and matched target preference is exactly chance. The frozen partition stays sealed and no writer is trained.

a mean margin of 0.747237. Thus candidate visibility is intact, but V26's history state does not expose a transferable next-glyph relation to this deterministic nonlinear decoder.

The resulting V27 direction is therefore stricter than replacing an output head. A trainable image retina and causal context field must be optimized jointly with deterministic candidate compatibility under matched alternatives. The primary score remains image-to-image and vocabulary-free; an appearance-only chance arm, raw-retina identity control, prefix shuffle, unigram, and symbolic bigram remain required controls. Stochastic form generation follows only after this relation passes.

Frozen V26 state does not expose next-glyph binding

Post-hoc localization diagnostic | development only | frozen split sealed | not model evidence



Cross-font paired accuracy

Vertical marker: chance. The positive control uses the same candidate images and frozen retina.



Diagnosis

History changes numerically, but a deterministic nonlinear scorer cannot decode a transferable next-glyph relation from it. Candidate visibility is not the bottleneck.

History margin 0.000108 | Fused margin 0.000371 | Retina margin 0.747237

V27 decision: jointly train the image context encoder and candidate-conditioned compatibility objective before attaching a stochastic visual writer.

One train-pair pass | 93.46 s | 2.482 GiB peak on one RTX 4090

Figure 15: Post-hoc frozen-V26 localization. The raw retina nearly perfectly distinguishes paired target images across unseen fonts, while deterministic candidate scorers over frozen history and fused states remain at chance. This is a development diagnostic, not preregistered model evidence.

5.10 Joint Visual Compatibility V27: visible candidates, failed relation

V27 executes that joint test without changing the corpus, record partitions, font separation, 64-cell horizon, or student boundary. Its online retina initializes from V16 and is optimized at one tenth the main learning rate. An eight-layer width-384 causal rotary field maps the ordered context images to a normalized query. An exponential-moving-average retina and residual image projector map an arbitrary candidate image to a normalized key:

$$q(X) = \text{normalize } Q(F(R_o(X))_{64}), \quad k(Y) = \text{normalize } G_t(R_t(Y)),$$

$$s(X, Y) = \exp(\alpha) q(X)^\top k(Y), \quad \exp(\alpha) \leq 100.$$

The candidate collection is not part of the model or checkpoint. At deployed inference the scorer accepts any floating candidate image. Every paired training view independently permutes its two candidate images, and the evaluator undoes only that visible-array permutation. This makes fixed diagonal position unusable as an identity channel.

The fixed objective combines multi-positive cross-font natural-image contrast, symmetric row-and-column assignment on suffix-matched pairs, cross-font candidate identity, and VICReg variance/covariance regularization:

$$\mathcal{L} = \mathcal{L}_{\text{natural}} + 2\mathcal{L}_{\text{pair}} + 0.5\mathcal{L}_{\text{identity}} + 0.05\mathcal{L}_{\text{VC}}.$$

No vocabulary classification, symbolic target, reconstruction decoder, stochastic particle head, or writer is trained. The complete model has 18,599,553 parameters, of which 17,118,401 are trainable. The single fixed run performs 8,000 BF16 AdamW updates. Training and the complete development audit take 2,347.47 seconds on one RTX 4090 with 2.267916 GiB peak allocated CUDA memory. The protocol SHA-256 is 0c386dd2...cb223310; the final checkpoint SHA-256 is 49dc07f2...1e77e539. The frozen partition is never rendered.

The natural audit uses 2,048 development windows and the same evaluator-only 1,024-image bank for every branch. Table 12 reports the fixed language comparisons.

V27 natural development measure	Measured	Required	Result
Full-context top-1 / top-5	0.016113 / 0.023926	diagnostic	–
Last-only / suffix-4 top-1	0.001953 / 0.008301	diagnostic	–
Shuffled-prefix top-1	0.013184	diagnostic	–
Image-unigram top-1	0.020508	control	below
Symbolic-bigram top-1	0.125977	benchmark	below
Full minus unigram top-1	−0.004395	> 0.03	fail
Full minus bigram top-1	−0.109863	> 0.01	fail
Full minus bigram log probability	−1.856770	> 0.05	fail
Full minus suffix-4 log probability	+0.059248	> 0.03	pass
Full minus shuffled log probability	+0.002726	> 0.03	fail
Learned 1,024-bank cross-font identity	0.948730	≥ 0.99	fail
Student boundary / peak CUDA	clean / 2.267916 GiB	clean / < 18	pass

Table 12: V27 fixed natural-Chinese development audit. Full context improves target log probability over the four-cell suffix, but the advantage almost vanishes when the same prefix images are shuffled. The model remains below image unigram and symbolic bigram controls.

Only one of five natural-language gates passes. Full target log probability is −6.876778, versus −6.936025 for suffix-4 and −6.879503 after shuffling cells 1–60. Thus the apparent context gain is

not attributable to the order of the earlier writing. Full top-1 is also lower than both frequency controls.

The primary causal audit contains 512 cross-record suffix-4 pairs in two unseen development fonts, yielding 1,024 assignments and 2,048 scored arms. Table 13 reports the fixed mechanism controls.

V27 suffix-4 pair measure	Measured	Required	Result
Suffix pixel equality	1.000000	exact	pass
Candidate permutation score error	0.000000	$< 10^{-5}$	pass
Candidate permutation agreement	1.000000	exact	pass
Raw V16 two-candidate identity	0.999512	≥ 0.99	pass
Learned 1,024-bank identity	0.948730	≥ 0.99	fail
Full-context arm accuracy	0.507080	> 0.65	fail
Full both-correct rate	0.088867	> 0.40	fail
Last-only / suffix-4 accuracy	0.500000 / 0.500000	exact chance	pass
Shuffled-prefix arm accuracy	0.505615	control	near chance
Full minus suffix-4 accuracy	+0.007080	> 0.15	fail
Full minus shuffled accuracy	+0.001465	> 0.05	fail
Full minus shuffled margin	+0.002265	> 0.02	fail

Table 13: V27 pixel-identical suffix intervention. Candidate visibility, permutation equivariance, suffix equality, and chance controls pass. The learned context/candidate relation remains at chance and does not materially exceed a suffix-preserving prefix shuffle.

The raw V16 retina reaches 0.999512 cross-font target identity with 0.736124 mean cosine margin. Candidate pixels and the evaluator are therefore adequate. This is a two-candidate pair control. The learned identity gate instead uses 1,024 candidates, reaches 0.948730, and therefore cannot be subtracted from the raw number as a before/after change. Full assignment remains only 0.1465 percentage point above the shuffled-prefix arm. Seven of thirteen mechanism gates pass, but all five context-binding gates and the learned full-bank identity gate fail.

V27 rejects this global query/key compatibility mechanism, not image-native language modeling in general. It also removes a tempting explanation for V26: the main failure is not that arbitrary candidate images are invisible to the retina. The next controlled experiment should preserve the raw retinal geometry exactly and supervise dense, explicitly ordered future visual fields rather than compressing all language evidence into one global dot product. The authoritative representation remains an ordered $N \times 1 \times 32 \times 32$ stream of full glyph images. A reversible 2D fold may expose parallel spatial computation, but must invert exactly and pass an order intervention. A writer, longer lattice, geometric depth, and motion remain unauthorized until ordered context beats suffix, shuffle, unigram, and bigram controls.

V28 executes this dense-future test below without changing the corpus, partitions, student boundary, or frozen-data rule.

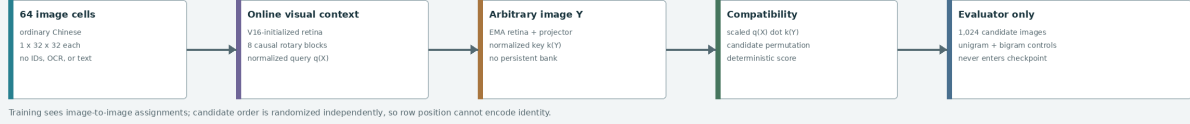
5.11 Dense Visual Future Energy V28: order signal without binding

V28 consumes an authoritative ordered volume $X \in \mathbb{R}^{64 \times 1 \times 32 \times 32}$ and predicts future writing images at offsets $d \in \{1, 2, 4\}$. The fixed V16 retina is never updated. For image x , a residual semantic adapter, initialized as the identity, and its stop-gradient EMA copy produce

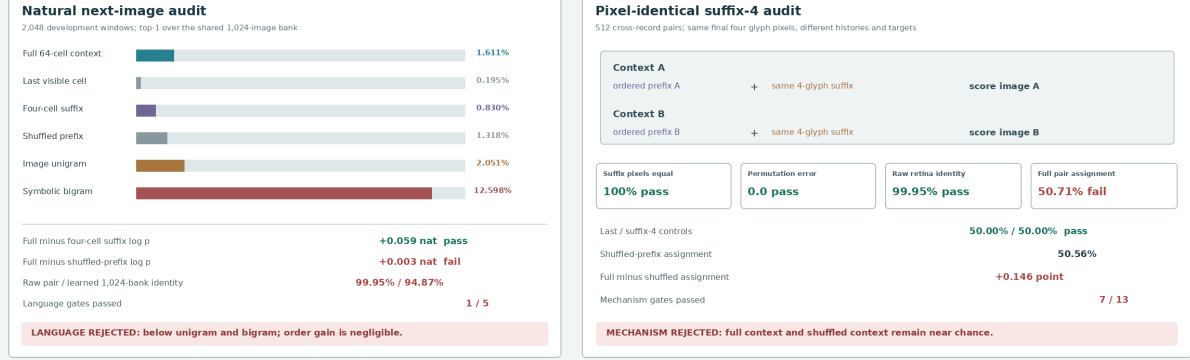
$$r(x) = \text{normalize } R_0(x), \quad z(x) = \text{normalize } A_\theta(r(x)), \quad \bar{z}(x) = \text{EMA}(z(x)).$$

V27: candidate images remain visible, but visual compatibility is not language

18.60M total parameters | 8,000 fixed BF16 updates | one RTX 4090 | 2.268 GiB peak | frozen split sealed



Training sees image-to-image assignments; candidate order is randomized independently, so row position cannot encode identity.



What the fixed experiment establishes

The image-only scorer is permutation-equivariant, the exact suffix intervention holds, and raw candidate visibility is 99.95%. The complete audit runs in 39.1 minutes with 2.268 GiB peak allocated VRAM; frozen writing stays sealed.

What V27 falsifies

Jointly adapting one global context query and one candidate-image key does not learn useful ordered next-writing compatibility. The separate learned 1,024-bank identity gate is 94.87%; full context is only 0.15 point above a prefix shuffle.

Next controlled question

Preserve the raw retina exactly and train dense, explicitly ordered future-field prediction over the full glyph-image stream. Require a causal advantage over suffix and shuffled history before adding a writer, longer lattice, depth, or motion.

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Figure 16: Measured V27 development result generated directly from the fixed audit. The image-only boundary, candidate permutation, exact suffix intervention, and raw-retina identity controls pass. Jointly learned visual compatibility remains near chance, misses its separate full-bank identity gate, and does not use prefix order. The frozen partition stays sealed and no writer is trained.

An eight-layer width-384 causal rotary field reads the concatenated raw and semantic states. At each selected position and horizon it emits four normalized raw hypotheses, four normalized semantic hypotheses, and mixture logits:

$$H_{1:t} = F([r(x_1); z(x_1)], \dots, [r(x_t); z(x_t)]),$$

$$\{q_{d,k}^r, q_{d,k}^z, \pi_{d,k}\}_{k=1}^4 = P_d(H_t).$$

The score of an arbitrary candidate image Y is continuous and contains no vocabulary table:

$$s_d(X, Y) = \log \sum_{k=1}^4 \pi_{d,k} \exp\left(\tau_r q_{d,k}^r r(Y) + \tau_z q_{d,k}^z \bar{z}(Y)\right).$$

Each natural microbatch samples 16 stratified causal positions, including the endpoint, and uses opposite-font targets at horizons 1, 2, and 4. The fixed objective is

$$\mathcal{L} = \mathcal{L}_{\text{dense}} + 0.25\mathcal{L}_{\text{ES}} + 0.5\mathcal{L}_{\text{id}} + 2\mathcal{L}_{\text{pair}} + \mathcal{L}_{\text{natural-order}} + \mathcal{L}_{\text{pair-order}}.$$

Here $\mathcal{L}_{\text{dense}}$ is weighted multi-positive cross-font contrast, $\mathcal{L}_{\text{ES}}$ is a strictly proper empirical energy score over the continuous raw hypotheses, and $\mathcal{L}_{\text{id}}$ trains the semantic route across fonts. The final

three terms use identical candidate pixels under symmetric suffix-pair assignment and under explicit full, suffix-only, and suffix-preserving shuffled-prefix interventions. Temporary equality groups are derived from canonical pixels by the host loss; they are not model inputs or inference state.

The complete model has 17,859,142 parameters, of which 16,377,990 are trainable. Its single fixed run performs 10,000 BF16 AdamW updates and the full development audit in 7,134.82 seconds (118.91 minutes) on one RTX 4090, with 1.144400 GiB peak allocated CUDA memory. The protocol SHA-256 is `b8e515d2...a7780f7f`, the final checkpoint SHA-256 is `22503464...2c7f64f`, and the audit SHA-256 is `2cb73707...8f0fb3d`. The frozen partition is never rendered.

The natural audit uses 2,048 development windows and one fixed evaluator-only 1,024-image bank. Table 14 reports the result.

V28 natural development measure	Measured	Required	Result
Full-context top-1 / top-5	0.014160 / 0.035156	diagnostic	–
Last-only / suffix-4 top-1	0.004395 / 0.015625	diagnostic	–
Shuffled-prefix top-1	0.002441	diagnostic	–
Image-unigram top-1	0.018555	control	below
Symbolic-bigram / trigram top-1	0.131348 / 0.193359	controls	below
Full minus unigram top-1	−0.004395	> 0.03	fail
Full minus bigram top-1	−0.117188	> 0.01	fail
Full minus bigram log probability	−1.801083	> 0.05	fail
Full minus suffix-4 log probability	+0.030369	> 0.03	pass
Full minus shuffled log probability	+0.215112	> 0.03	pass
Raw / EMA 1,024-way identity	0.920410 / 0.964355	EMA $\geq$ 0.95	pass
EMA minus raw identity	+0.043945	$\geq$ 0.02	pass
Student boundary / peak CUDA	clean / 1.144400 GiB	clean / < 18	pass

Table 14: V28 fixed natural-Chinese development audit. Full context changes target probability relative to suffix and shuffled controls, and the semantic route improves same-scope identity. Correct future-image rank remains below the unigram, bigram, and trigram controls.

Full target log probability is -6.829707 , versus -6.860076 for suffix-4 and -7.044819 after shuffling cells 1–60. This is an attributable order effect in probability, but suffix-4 top-1 is slightly higher than full-context top-1 and the symbolic bigram is more than nine times as accurate. Four of six language gates fail.

The primary causal audit uses 512 cross-record context pairs whose final four glyph images are bitwise equal but whose earlier histories and targets differ. Candidate order is independently permuted in two unseen development fonts. Table 15 reports the fixed mechanism controls.

V28 is a clean negative result. It establishes that the $N \times 1 \times 32 \times 32$ visual-time interface, frozen-retina semantic adaptation, dense multi-horizon prediction, and continuous candidate scoring are executable within 1.15 GiB peak allocated memory. It does not establish a useful language model or a capability-normalized efficiency advantage. The failure is now narrower: ordered prefix information changes scores, but the training objective does not distribute that evidence reliably enough to bind each context to its correct future image.

V29 executes that candidate-conditioned incremental-evidence test below without changing the corpus, partitions, student boundary, or frozen-data rule.

5.12 Conditional Visual Density Ratio V29: order without assignment

V29 retains the authoritative image stream $X \in [0, 1]^{B \times N \times 1 \times 32 \times 32}$, where N is reading time. It loads the V28 context input, eight causal blocks, context normalization, frozen V16 retina, and both

V28 suffix-4 pair measure	Measured	Required	Result
Suffix pixel equality	1.000000	exact	pass
Permutation error / agreement	0.000000 / 1.000000	exact	pass
Raw V16 two-candidate identity	0.999512	diagnostic	–
Full-context arm accuracy	0.495605	> 0.65	fail
Full both-correct rate	0.096680	> 0.40	fail
Last-only / suffix-4 accuracy	0.500000 / 0.500000	exact chance	pass
Shuffled-prefix arm accuracy	0.499512	control	chance
Full minus suffix-4 accuracy	−0.004395	> 0.15	fail
Full minus shuffled accuracy	−0.003906	> 0.05	fail
Full minus shuffled mean margin	+0.022070	> 0.02	pass

Table 15: V28 pixel-identical suffix intervention. The full score has a larger mean margin than the shuffled score on a subset of examples, but this movement does not yield reliable correct rank or context-to-target binding.

frozen semantic adapters. V28 future heads are discarded. The causal field returns one retained state per visible image:

$$H_T = C_\theta([r(x_1); z(x_1)], \dots, [r(x_T); z(x_T)]) \in \mathbb{R}^{T \times 384}.$$

An arbitrary candidate image y forms a continuous query $q_0 = W_y[r(y); \bar{z}(y)]$. Two six-head cross-attention layers update this query from all context states. The scalar relation head receives the updated query, its displacement, and its elementwise interaction with the original query:

$$q_{l+1} = \text{EvidenceLayer}_l(q_l, H_T), \quad \rho_\theta(H_T, y) = M_\theta([q_2; q_2 - q_0; q_2 \odot q_0]).$$

There is no candidate-ID embedding or bank-sized output layer.

For prefix P , exact four-image suffix S , and candidate y , define

$$F(P, S, y) = \rho_\theta(H(P, S), y), \quad B(S, y) = \rho_\theta(H(S), y),$$

$$G(P, S, y) = F(P, S, y) - B(S, y).$$

Rows of G are centered over candidates. Natural training contrasts all 1,024 host-side candidate images for full, suffix, and incremental scores and adds an ordered-versus-shuffled increment margin. Exact-suffix pairs add symmetric two-by-two full and incremental assignment, per-row positive margin, and per-row ordered-versus-shuffled margin. Candidate images are encoded once by the frozen image path, but their bank is neither serialized nor required at inference. The student receives no bank index, source string, token, Unicode ID, character ID, OCR result, symbolic probability, or external-model state.

The complete model has 20,080,961 parameters, of which 18,451,585 are trainable. Its single fixed run performs 8,000 BF16 AdamW updates and the full development audit in 4,956.14 seconds (82.60 minutes) on one RTX 4090, with 3.356465 GiB peak allocated CUDA memory. The protocol SHA-256 is `4cb17b79...9ffad0f`, the final checkpoint SHA-256 is `a8ec9919...d508b8a`, and the audit SHA-256 is `16645844...4a5108`. Frozen images are never instantiated.

The fixed natural audit uses 2,048 development windows and one evaluator-only 1,024-image bank. Table 16 reports the result.

The primary causal audit uses 512 cross-record pairs whose final four writing images are bitwise equal while earlier histories and next images differ. Candidate order is independently permuted in two unseen development fonts. Table 17 reports the fixed mechanism controls.

V28: dense visual futures learn form and order, but fail language choice

17.86M total / 16.38M trainable | 10,000 BF16 updates | one RTX 4090 | 1.144 GiB peak | frozen split sealed

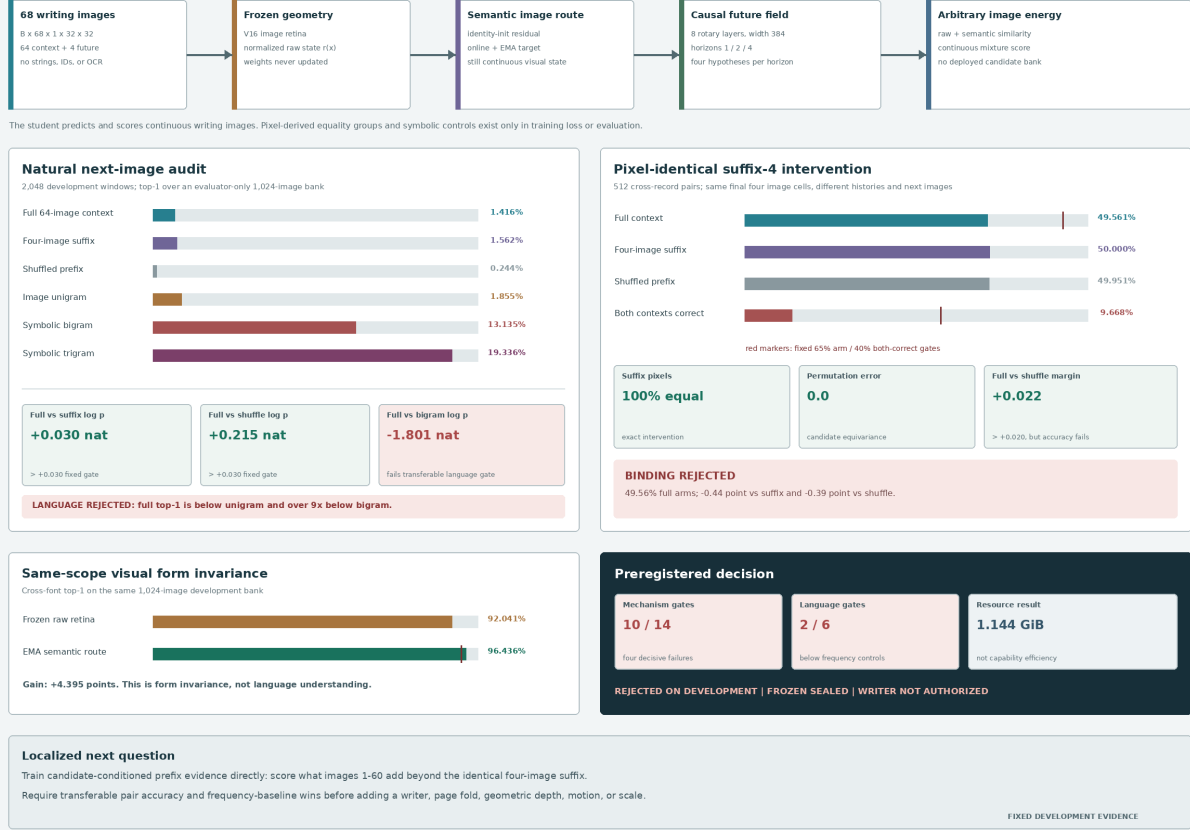


Figure 17: Measured V28 development result generated directly from the fixed audit. Frozen raw geometry and the learned semantic route preserve candidate visibility, and ordered history affects probability and mean margin. Natural selection remains below frequency controls and matched pair assignment stays at chance. Ten of fourteen mechanism gates and two of six language gates pass, so frozen evaluation and writer training remain unauthorized.

The exact suffix exposes a structural limit of the subtraction. Both context rows share one baseline vector (b_1, b_2) . For the diagonal assignment, the mean incremental margin is

$$\begin{aligned} & \frac{1}{2}[(G_{11} - G_{12}) + (G_{22} - G_{21})] \\ &= \frac{1}{2}[(F_{11} - b_1 - F_{12} + b_2) + (F_{22} - b_2 - F_{21} + b_1)] \\ &= \frac{1}{2}[(F_{11} - F_{12}) + (F_{22} - F_{21})]. \end{aligned}$$

The suffix terms cancel exactly, and row centering leaves within-row margins unchanged. Subtraction can redistribute margin between rows, but cannot improve the aggregate assignment margin beyond the full critic. The fixed audit confirms that full and incremental mean margins are both 0.005103277.

V29 is a clean negative result. It rejects this candidate-query critic and density-ratio loss, not image-native language modeling or the ordered visual cell interface. The failure is now more specific: a global scalar critic can learn whether a history resembles natural order without learning which candidate image belongs to which history. A bounded successor should predict a continuous next-

V29 natural development measure	Measured	Required	Result
Full-context top-1 / top-5	0.023438 / 0.063965	top-1 ≥ 0.15	fail
Suffix-4 / shuffled top-1	0.013672 / 0.011719	diagnostic	–
Increment / shuffled increment top-1	0.023438 / 0.008789	diagnostic	–
Image-unigram top-1	0.013672	control	full above
Symbolic-bigram / trigram top-1	0.138672 / 0.202637	controls	full below
Full minus unigram top-1	+0.009766	> 0.03	fail
Full minus bigram top-1	−0.115234	> 0.01	fail
Full minus bigram log probability	−1.516622	> 0.05	fail
Full minus suffix-4 log probability	+0.179446	> 0.03	pass
Full minus shuffled log probability	+3.102479	> 0.03	pass
Raw / frozen-semantic identity	0.920410 / 0.964355	semantic ≥ 0.95	pass
Boundary / bank absence / peak CUDA	clean / clean / 3.356465 GiB	required	pass

Table 16: V29 fixed natural-Chinese development audit. Earlier ordered images strongly affect target probability, but full 1,024-way ranking remains far below a symbolic bigram. Order detection is not conditional visual choice.

V29 suffix-4 pair measure	Measured	Required	Result
Suffix pixels / score rows	1.000000 / 1.000000	exact	pass
Permutation error / agreement	0.000000 / 1.000000	exact	pass
Raw two-candidate identity	0.999512	≥ 0.99	pass
Full arm / both-correct	0.497314 / 0.038086	> 0.65 / > 0.40	fail
Suffix-only arm accuracy	0.500000	exact chance	pass
Shuffled-prefix arm accuracy	0.501221	control	chance
Increment arm / both-correct	0.507080 / 0.089844	> 0.65 / > 0.40	fail
Shuffled-increment arm accuracy	0.495605	control	chance
Increment minus shuffled accuracy	+0.011475	> 0.10	fail
Increment minus shuffled margin	+0.003167	> 0.05	fail

Table 17: V29 pixel-identical suffix intervention. Perception, exactness, and candidate equivariance pass. Full and incremental candidate assignment remain at chance and fail every learned binding threshold.

image patch field from context, encode an arbitrary candidate into the same field, and compare corresponding patches before scalar aggregation. A parameter- and compute-matched bilinear visual scorer should control whether spatial interaction, rather than capacity, provides any gain. The existing suffix, shuffle, frequency, permutation, identity, and boundary controls must pass before a writer, page fold, geometric depth, motion, or larger model is authorized.

5.13 Spatial Visual Next-Field V30: local sensitivity without language

V30 changes the predicted random variable rather than the scalar-score baseline. Both arms retain the V29 causal context field and frozen visual perception, discard the candidate-query critic, and emit a candidate-independent continuous next-image field:

$$P_{\theta}(X) \in \mathbb{R}^{16 \times 192}.$$

V29: visual order signal grows, but candidate binding remains at chance

20.08M total / 18.45M trainable | 8,000 BF16 updates | one RTX 4090 | 3.356 GiB peak | frozen split sealed

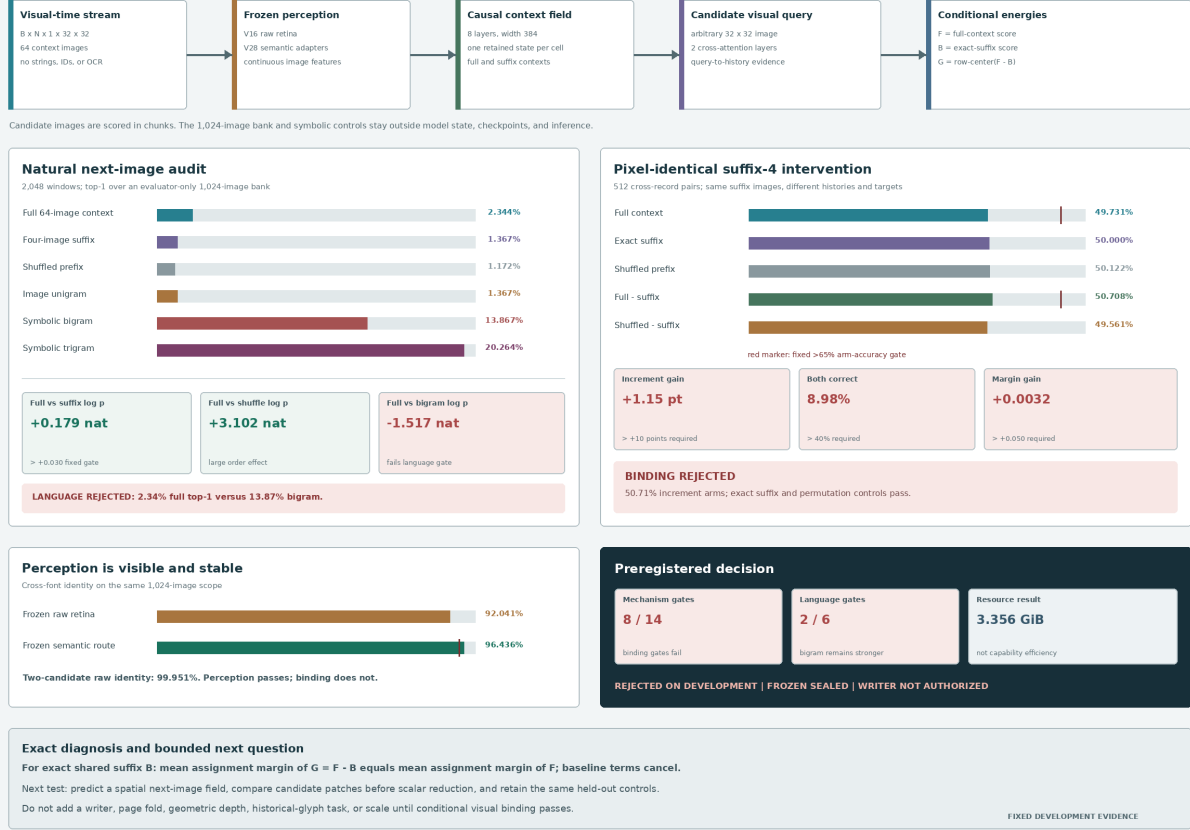


Figure 18: Measured V29 development result generated directly from the fixed audit. The critic detects a large natural order effect while exact-suffix candidate binding stays at chance. All perception, equality, permutation, leakage, and resource controls pass. Eight of fourteen mechanism gates and two of six language gates pass, so frozen evaluation and writer training remain unauthorized.

The 16 rows preserve the frozen retina’s 4×4 topology. For an arbitrary candidate image y , the spatial arm compares corresponding normalized retinal cells before reduction,

$$s_{sp}(X, y) = \tau \frac{1}{16} \sum_{p=1}^{16} P_{\theta,p}(X)^\top u_p(y).$$

The matched control repeats the candidate’s frozen global-semantic vector over the same rows,

$$s_{gl}(X, y) = \tau \frac{1}{16} \sum_{p=1}^{16} P_{\theta,p}(X)^\top z(y).$$

It preserves candidate visibility, trainable capacity, score reduction, and compute topology while removing local correspondence. A 180-degree reversal of the candidate cells intervenes on the spatial route and leaves the global route exactly invariant.

The arms have 18,641,153 total and 17,011,777 trainable parameters. They start from tensors with the same keys, shapes, dtypes, values, and SHA-256 (507d875d...d9547ca8e). Each independently completes 8,000 finite BF16 AdamW updates on one RTX 4090. The spatial and global

runs take 2,693.27 and 3,789.99 seconds and peak at 1.595290 and 1.597071 GiB allocated memory. The joint audit takes 39.33 seconds. These are resource measurements for a rejected mechanism, not capability-normalized efficiency claims. The protocol, spatial checkpoint, global checkpoint, and audit SHA-256 values begin 81d2b2af, 11a3a7e9, 66378d4b, and 2d0a3a08, respectively.

Table 18 reports the fixed 2,048-window natural audit. Both routes detect ordered context, but the global route is stronger and both remain far below the symbolic bigram.

V30 natural development measure	Spatial	Global	Spatial requirement
Full-context top-1	0.012695	0.024902	≥ 0.15 ; fail
Suffix-4 / shuffled top-1	0.006348 / 0.000488	0.015625 / 0.001465	diagnostic
Patch-reversed top-1	0.000488	0.024902	diagnostic
Image-unigram top-1	0.013184	0.013184	spatial below
Symbolic-bigram top-1	0.117188	0.117188	spatial below
Symbolic-trigram top-1	0.200195	0.200195	diagnostic
Full minus shuffled log probability	+0.315340	+0.650060	> 0.03 ; pass
Full minus patch-reversed log probability	+0.015412	0.000000	> 0.05 ; fail
Spatial minus global top-1	−0.012207		> 0.01 ; fail
Spatial minus global log probability	−0.190402		> 0.05 ; fail
Cross-font candidate identity	0.963379	0.964355	≥ 0.95 ; pass

Table 18: V30 fixed natural-Chinese development audit. Patch alignment affects the spatial rank, but its target-probability effect misses the causal gate and the parameter-identical global control is stronger.

The 512-pair intervention keeps the final four context images bitwise equal while changing earlier history and the next image. Table 19 shows that neither route binds the correct candidate to the earlier context.

V30 suffix-4 pair measure	Spatial	Global	Spatial requirement
Suffix pixels / score-row error	1.000000 / 0	1.000000 / 0	exact; pass
Full arm accuracy	0.500488	0.505859	> 0.65 ; fail
Full both-correct rate	0.050781	0.081055	> 0.40 ; fail
Suffix-only arm accuracy	0.500000	0.500000	exact chance
Shuffled-prefix arm accuracy	0.492920	0.491943	control
Patch-reversed arm accuracy	0.496582	0.505859	control
Full minus shuffled accuracy	+0.007568	+0.013916	> 0.10 ; fail
Full minus shuffled mean margin	+0.010235	+0.034522	> 0.05 ; fail
Full minus patch-reversed accuracy	+0.003906	0.000000	> 0.05 ; fail
Candidate-column error / agreement	0 / 1.000000	0 / 1.000000	exact; pass
Spatial minus global full accuracy	−0.005371		> 0.05 ; fail

Table 19: V30 pixel-identical suffix intervention. Candidate pixels, exact suffix equality, and column equivariance are controlled. Local patch alignment does not produce reliable context-to-candidate assignment.

The field’s sensitivity is real but insufficient. Reversal changes spatial scores by as much as 3.297, yet it improves target log probability by only 0.0154 nat and pair correctness by 0.391 percentage point. A single deterministic field with bilinear candidate energy is encouraged toward an average next retinal state. Natural next writing is multimodal; the fixed contrastive and pair losses do not make this one field represent the required conditional alternatives. This interpretation does not reject all spatial continuous distributions. It rejects this field, decoder, and objective on

the fixed benchmark.

V30 passes 12/18 spatial common gates, 12/12 global integrity gates, 5/9 matched-arm gates, and 0/8 spatial language gates. Frozen images remain uninstantiated. This failure motivates the separately preregistered V31 multimodal continuous-field test below; it does not authorize a writer, page fold, depth axis, or historical-glyph task.

V30: aligned spatial fields change scores, but do not bind next writing

Two parameter-identical 18.64M models | 8,000 BF16 updates each | one RTX 4090 | 1.60 GiB peak per arm | frozen sealed

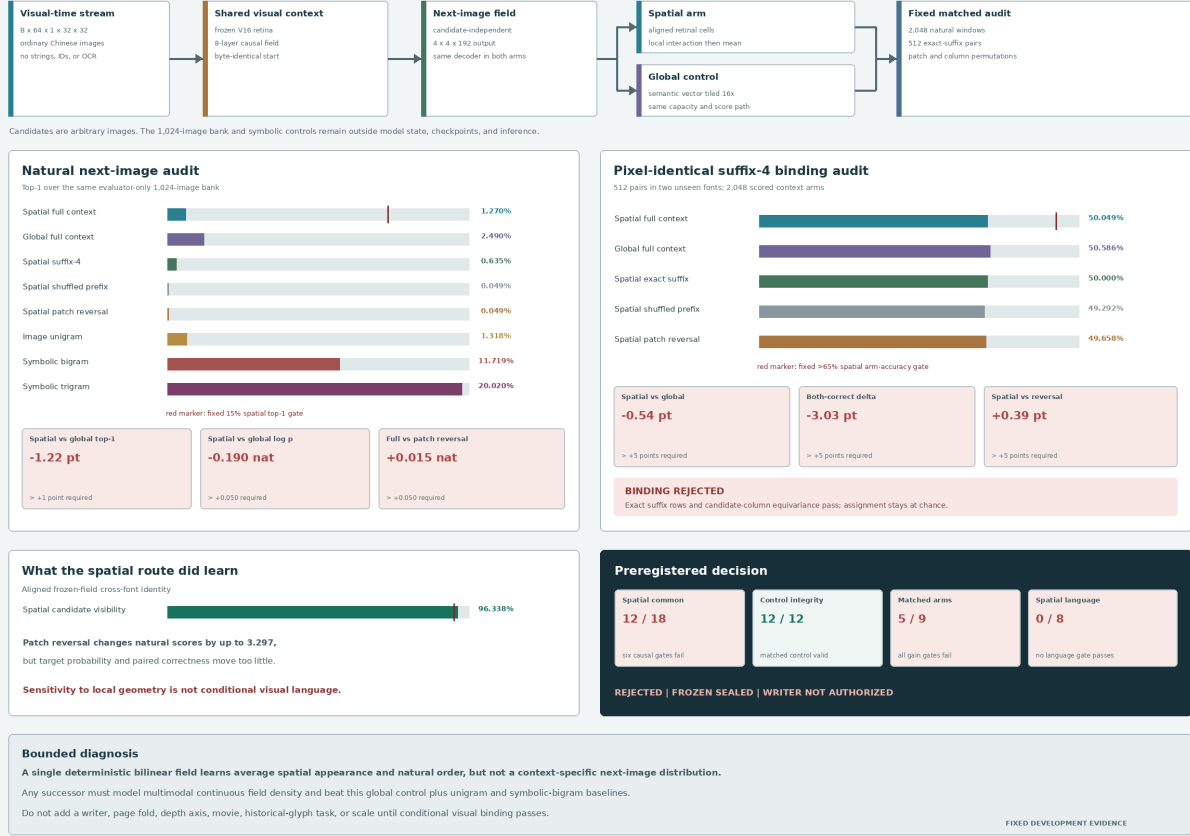


Figure 19: Measured V30 matched-arm result generated directly from the fixed audit. The spatial route is sensitive to local candidate alignment but is worse than the parameter-identical global route on natural top-1, target log probability, pair assignment, and pair both-correct. It passes 12/18 common gates and 0/8 language gates; the control passes all 12 integrity gates.

5.14 Conditional Visual Field Flow V31: context response without language

V31 executes the multimodal continuous-field test prescribed by V30. The candidate-independent causal reader remains image-only, but the deterministic field decoder is replaced by a context-conditioned velocity field. For normalized target retinal field Y , coherent Gaussian base field E , context state h , and $t \sim \mathcal{U}(0, 1)$, the interpolation and target velocity are

$$Z_t = (1 - t)E + tY, \quad u^*(Z_t, t, h) = Y - E.$$

The learned field minimizes the conditional flow-matching objective

$$\mathcal{L}_{\text{flow}}(\theta) = \mathbb{E}_{h, Y, E, t} \left[\|v_\theta(Z_t, t, h) - (Y - E)\|_2^2 \right].$$

One base vector is tiled coherently over the full retinal field; independent per-patch noise is forbidden. Thus one draw denotes one global visual alternative rather than a collage of independently sampled patches. The spatial arm predicts a 16×192 field. The matched global arm predicts one semantic vector tiled across the same 16 cells.

The audit reports both a path score and autonomous generation. For eight fixed probe times t_k , evaluator-only candidate ranking uses the negative velocity residual along each coherent base-to-candidate path,

$$s_\theta(Y \mid h, E) = -\frac{1}{K} \sum_{k=1}^K \|v_\theta((1 - t_k)E + t_k Y, t_k, h) - (Y - E)\|_2^2.$$

Autonomous fields integrate

$$\frac{dZ_t}{dt} = v_\theta(Z_t, t, h), \quad Z_0 = E,$$

with eight-step Heun integration. The 1,024-image bank is evaluator-only: it is absent from student inputs, parameters, checkpoints, and sampling. The displayed nearest glyph to a generated field is therefore a diagnostic proxy, not the model’s generated pixel image.

Both arms contain 18,736,577 total and 17,107,201 trainable parameters. They start from byte-identical initialized states (0682269c...2ad3d3) and each completes exactly 10,000 finite BF16 updates. Spatial and global training take 4,217.57 and 4,470.01 seconds and peak at 0.99670 and 0.99661 GiB allocated CUDA memory. These are resource receipts for a rejected mechanism, not capability-normalized efficiency claims. The protocol, spatial checkpoint, global checkpoint, development audit, and comparison SHA-256 values begin 92b6f709, 9808e996, 485011a4, 530c21d3, and 2e97b732, respectively.

Table 20 reports the fixed 2,048-window natural audit. The spatial route passes its order and local-layout log-probability interventions, and autonomous samples pass diversity and condition-sensitivity gates. Correct next-writing identity nevertheless falls below every useful baseline.

V31 natural development measure	Spatial	Global
Path full-context top-1	0.000977	0.018555
Path suffix-4 / shuffled top-1	0.001465 / 0.000000	0.013672 / 0.007812
Path spatial-permuted top-1	0.009277	0.018555
Autonomous full / shuffled top-1	0.001465 / 0.000000	0.005371 / 0.000000
Image-unigram top-1	0.016113	0.016113
Symbolic-bigram top-1	0.135254	0.135254
Symbolic-trigram top-1	0.209961	0.209961
Path full minus shuffled log probability	+0.125880	+0.131427
Path full minus spatial-permuted log probability	+0.052740	0.000000
Autonomous pairwise cosine distance	0.302390	0.933120
Same-noise full/shuffled displacement	5.736859	1.293999
Best-of-eight mean target rank	425.78	103.04
Cross-font candidate identity	0.963379	0.964355

Table 20: V31 fixed natural-Chinese development audit. Ordered and spatial interventions change the conditional field, but the spatial route identifies the correct next glyph in only two of 2,048 path rankings; autonomous nearest-bank matching identifies three of 2,048. Both are far below image-frequency and symbolic sequence baselines.

The 512-pair intervention changes earlier visual history and target while holding the final four context images pixel-identical. Table 21 shows that both path scoring and autonomous sampling remain at chance.

The spatial arm passes 14/19 common gates, the control passes 6/6 integrity gates, matched arms pass 4/8 gates, and spatial language and generation pass 0/10 gates. Frozen images remain uninstantiated; frozen evaluation and direct pixel writer training remain unauthorized. Figure 20 combines the architecture, measurements, and autonomous nearest-image diagnostic.

V31 suffix-4 pair measure	Spatial	Global
Suffix pixel equality / score-row max error	1.000000 / 0	1.000000 / 0
Path full-context arm accuracy	0.504883	0.514648
Path full both-correct rate	0.035156	0.078125
Path suffix-only arm accuracy	0.500000	0.500000
Path shuffled-prefix arm accuracy	0.492676	0.487793
Path full-minus-shuffle arm accuracy	0.012207	0.026855
Path full-minus-shuffle mean margin	0.005293	0.015111
Autonomous full-context arm accuracy	0.501953	0.505859
Autonomous full both-correct rate	0.008789	0.045898
Path spatial-permutation max score error	4.939047	0.000000

Table 21: V31 pixel-identical suffix intervention. Exact equality and candidate-column controls pass. Spatial flow responds to local permutation but does not bind earlier visual context to the correct target.

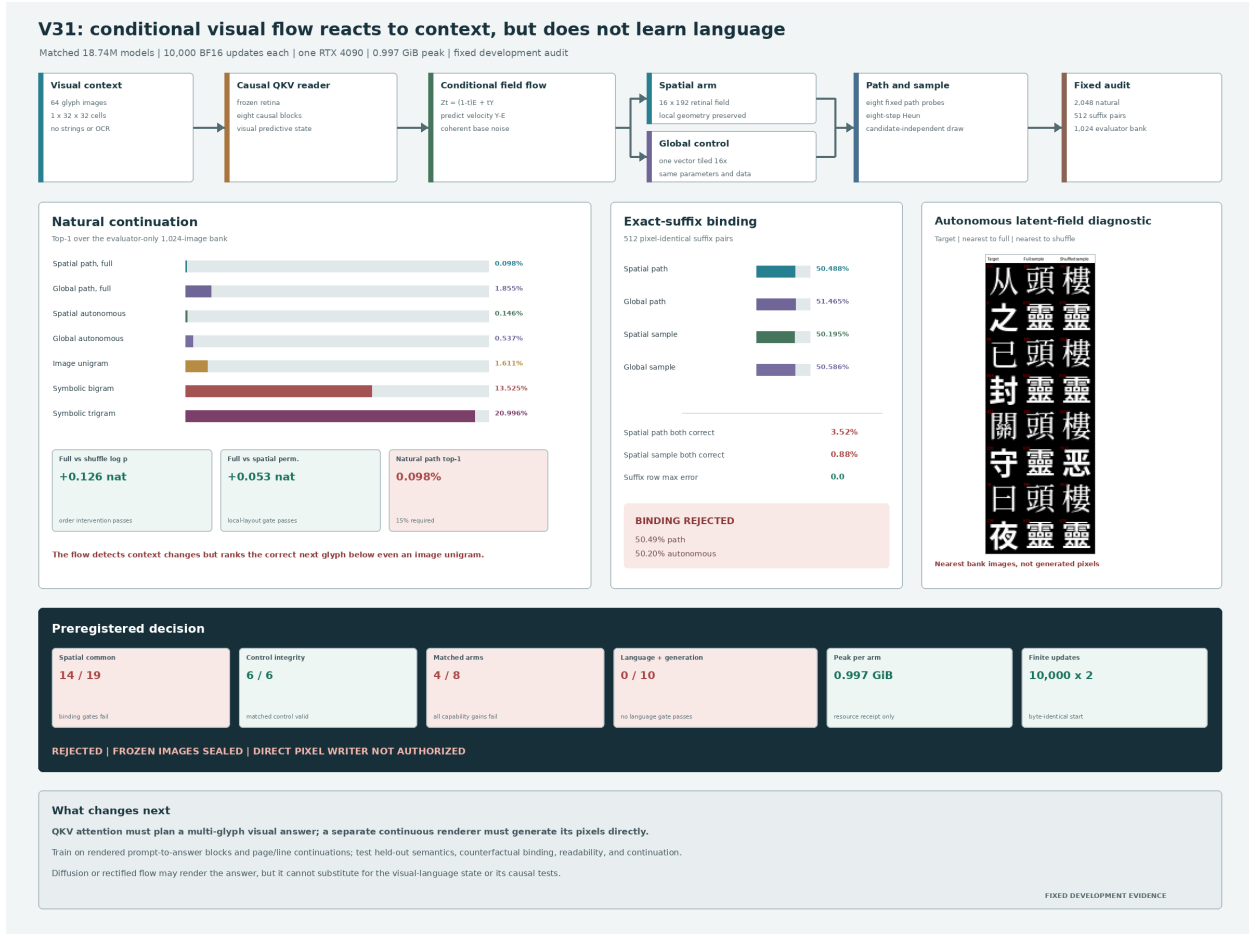


Figure 20: Measured V31 matched-arm result generated directly from the fixed audit. Conditional flow detects order, local layout, diversity, and context dependence while failing next-glyph identity and exact-suffix assignment. The embedded glyph columns are evaluator-nearest images to generated latent fields, not direct generated pixels.

V31 rejects one-glyph latent-field flow, not continuous visual generation in general. The next bounded proof changes both task and output variable. A causal QKV reader must form a visual predictive state from rendered prompts; a separate compact continuous head must directly render a multi-glyph answer block. A small reproducible curriculum should mix visual instruction-to-answer mappings with line/block continuation and hold out semantic compositions. Prompt counterfactuals, content swaps, paraphrases, generated-raster readability, and closed-loop continuation must all pass. Diffusion, rectified flow, or continuous autoregression may implement the renderer, but cannot substitute for causal language evidence. LocalLLM may author or audit fixed offline examples; it may not enter student batches, checkpoints, or inference. Open corpora provide the published benchmark, while local books enter only through a provenance-tracked preparation layer after this compact proof succeeds.

5.15 Continuous Glyph Representation Codec V34

V32–V33.1 showed that a language experiment is uninterpretable when the answer representation cannot preserve the target raster. V34 therefore isolates the visual interface before testing causality.



Figure 21: V31 autonomous latent-field diagnostic. Each row shows the target and the external evaluator’s nearest glyph under full and shuffled context. Repeated wrong modes expose generation failure. The model did not render these nearest glyph pixels.

For a binary writing patch $p \in \{0, 1\}^{1 \times 32 \times 32}$, the convolutional encoder and decoder are

$$z = \text{LN}(E_\phi(p)) \in \mathbb{R}^{768}, \quad \hat{p} = \sigma(D_\psi(z)).$$

The 7,423,361-parameter model uses four convolutional scales with channels 32, 64, 96, 192, a continuous normalized latent, and no VQ, codebook, character label, OCR target, or external model. Its objective is

$$\mathcal{L}_{\text{codec}} = \mathcal{L}_{\text{weighted BCE}} + 0.5 \mathcal{L}_{\text{Sobel}} + 0.5 \mathcal{L}_{\text{ink Dice}}.$$

Half of training patches receive independently sampled latent noise with standard deviation in $[0, 0.05]$.

The public stream contains 7,017 public-domain Chinese book records with different Noto CJK files for train, development, and sealed rendering. The historical stream contains 84,626 local SVG records partitioned by modern character family, so all known forms of one modern character remain in one split. This local corpus is neither copied into the repository nor treated as a redistribution

grant. V34 completes 6,000 BF16 updates in 321.31 seconds on one RTX 4090, processes 4,519,796 patches, and peaks at 2.38 GiB allocated memory.

V34 EMA measure	Development	Sealed
Rendered clean ink F1	0.997864	0.997793
Rendered clean edge F1	0.997270	0.997291
Rendered noisy-latent ink F1	0.997836	0.997749
Historical ink F1	0.998311	0.998068
Historical edge F1	0.997934	0.997747
Rendered OCR retention	1.002567	1.006088
Blank false-ink rate	0	0

Table 22: V34 development and once-opened sealed results. The sealed audit was authorized only after every development gate passed; every required sealed metric retained more than 97% of its development value. OCR is an evaluator, not a codec input or objective.

V34 is therefore **continuous-codec-qualified**. It establishes that a compact non-quantized interface can preserve held-font modern writing and held-family historical forms. It does not establish continuation, prompt following, factual knowledge, or autonomous language generation.

5.16 Causal Glyph Flow V35: complete raster loop, failed binding

Transferred foundation and student boundary. V35 tests whether a qualified visual interface and a transferred causal prior are sufficient for short Chinese continuation, copying, and instruction answering. The causal core is initialized from PIXAR’s released 12-layer, width-768 transformer [13]. This prior is external work. The local source revision is MIT, while the separately hosted weight archive states no weight license; the derived weight artifact is consequently not authorized for redistribution. The V34 EMA codec is this project’s frozen interface. At deployed inference, both foundations are packaged into one checkpoint and no PIXAR process, tokenizer, OCR engine, retrieval system, database, or language teacher is called. Thus *independent* denotes a self-contained visual runtime, not from-scratch provenance.

For visible patches $p_{\leq t}$, frozen codec E, D , aligned residual adapter A , and causal transformer T ,

$$z_t = E(p_t), \quad a_t = A(z_t), \quad h_t = T(a_{\leq t}).$$

The deterministic writer gives a normalized anchor

$$m_{t+1} = \text{LN}(M(h_t)).$$

The conditional writer instead learns a rectified flow. For target latent z , Gaussian base ϵ , and $\tau \sim \mathcal{U}(0, 1)$,

$$x_\tau = (1 - \tau)\epsilon + \tau z, \quad v^* = z - \epsilon, \\ \mathcal{L}_{\text{flow}} = \mathbb{E} \left[\|F_\theta(x_\tau, \tau, h_t) - v^*\|_2^2 \right].$$

The anchor supplies a one-pass route; the three-block width-512 flow head tests whether a conditional continuous distribution improves upon that route. A scalar visual-position head predicts when generation should stop.

The critical recurrence is visible rather than latent. Every predicted latent is decoded, thresholded into the emitted binary patch, and encoded again:

$$\tilde{p}_{t+1} = \mathbf{1}[\sigma(D(\hat{z}_{t+1})) \geq 0.5], \quad z_{t+1}^{\text{feedback}} = E(\tilde{p}_{t+1}).$$

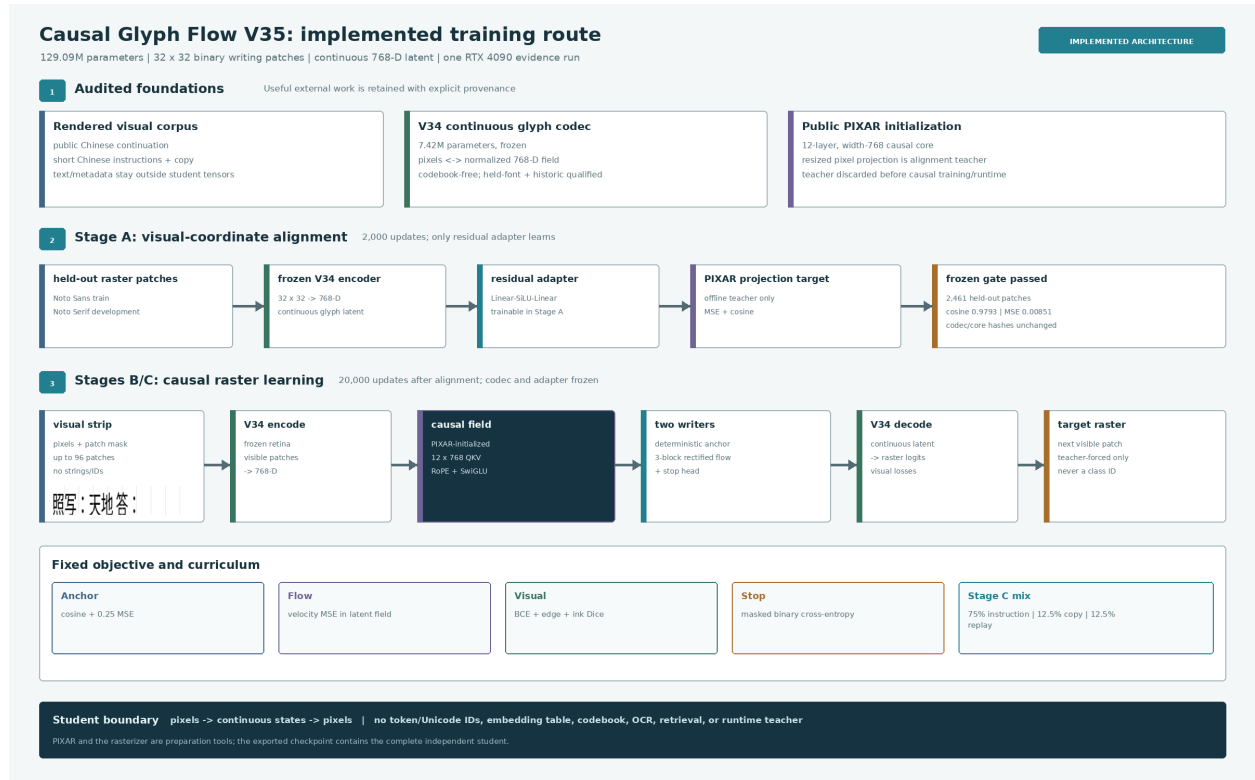


Figure 22: Implemented V35 training path. Stage A aligns the V34 latent interface to PIXAR’s visual coordinate system while both the codec and causal core are frozen. The alignment teacher is then removed. Stages B and C train the causal field and continuous writers from raster-only student batches. PIXAR initialization and offline text rendering are explicitly outside the deployed student boundary.

No unrendered writer state crosses this recurrent boundary. The fixed causal objective is

$$\begin{aligned}\mathcal{L}_{\text{total}} = & \mathcal{L}_{\text{flow}} + (1 - \cos(m, z) + 0.25\|m - z\|_2^2) \\ & + 0.25 \mathcal{L}_{\text{visual}} + 0.10 \mathcal{L}_{\text{stop}},\end{aligned}$$

where $\mathcal{L}_{\text{visual}}$ combines pixel BCE, Sobel edge error, and ink Dice error. Flow and decoded-visual losses use at most 128 sampled valid positions per microbatch; anchor and stop losses use every supervised position.

Stage A runs for 2,000 updates. It passes with adapter MSE 0.008510 and cosine 0.979348 on 2,461 held-out patches while the causal core and codec hashes remain unchanged. Stage B runs 8,000 updates on public continuation. Stage C runs 12,000 updates with 75% Chinese Alpaca instructions, 12.5% deterministic visual-copy tasks, and 12.5% public replay. Physical batch 8, eight-way accumulation, BF16, gradient checkpointing, EMA, and maximum context 96 keep the 129,092,738-parameter model at 2.899 GiB peak allocated memory. All 22,000 updates are finite and complete in 9,972.46 seconds.

Fixed development audit. EMA is the preregistered primary state. The anchor writer is selected because its fixed selection-set OCR character accuracy is 0.4241%, versus 0 for the eight-step flow writer; neither is useful. The raw-weight diagnostic also selects anchor at 0.4464%, versus 0.2232% for flow. Raw public teacher-forced ink and edge F1 are 0.3398 and 0.4760, effectively the same as EMA’s 0.3397 and 0.4754, so EMA selection does not conceal a capable raw route. The full audit evaluates every teacher-forced development record and autonomous correct, cyclically shuffled, blank, final-quarter, paraphrase, and copy-counterfactual conditions. Identical generation noise is reused across each intervention. OCR 5.3.4 with `chi_sim+chi_tra` is post-hoc only. In this audit, *readable* means only that OCR returned a nonempty string; it is not a human-readability claim.

V35 autonomous condition	Character accuracy	Nonempty OCR	Ink F1
Copy, correct	0.3125%	87.50%	0.09214
Copy, shuffled	0.1953%	87.50%	0.08645
Copy, blank	0	84.38%	0.07670
Public continuation, correct	0	71.88%	0.21265
Instruction, correct	0.1116%	75.00%	0.17346
Instruction, shuffled	0.5565%	75.00%	0.13709
Instruction, blank	0	100%	0.14888
Paraphrase, correct	0.3002%	87.10%	0.15300

Table 23: V35 autonomous EMA-anchor development results. All rows are nonblank, but nonempty OCR is not correct language. Correct instruction prompts perform worse than shuffled prompts, directly rejecting semantic binding.

The copy target-through-codec ceiling is 63.676%, already below its fixed 70% audit floor. Generated copy retains only 0.491% of that ceiling, versus the required 60%; correct prompts gain only 0.117 percentage point over shuffled and 0.313 point over blank prompts. Copy counterfactual target preference is 62.5%, below 75%, with zero mean character accuracy. The instruction ceiling passes at 78.858%, but autonomous correct-prompt accuracy is 0.1116%, against an 8% gate. It is 0.445 point worse than shuffled and only 0.112 point above blank, against required +2 and +3 point margins. The pixels change when prompts are shuffled or blanked, but the changes are not bound to the requested answer.

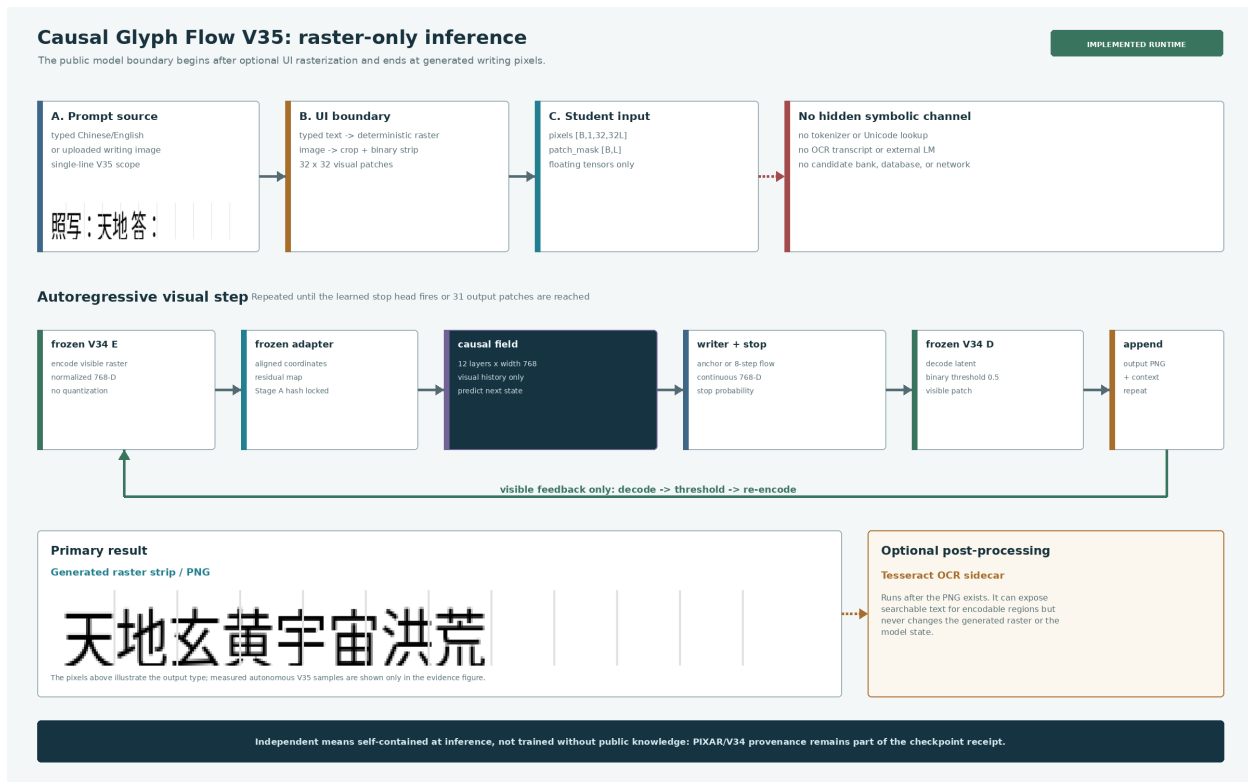


Figure 23: Implemented V35 inference path. A typed prompt is rasterized before the model or a writing image enters directly. The student sees pixels and a visual-position mask, generates raster patches through the selected continuous writer, and rereads only the thresholded visible result. OCR and coded text are optional post-hoc sidecars and cannot affect generation.

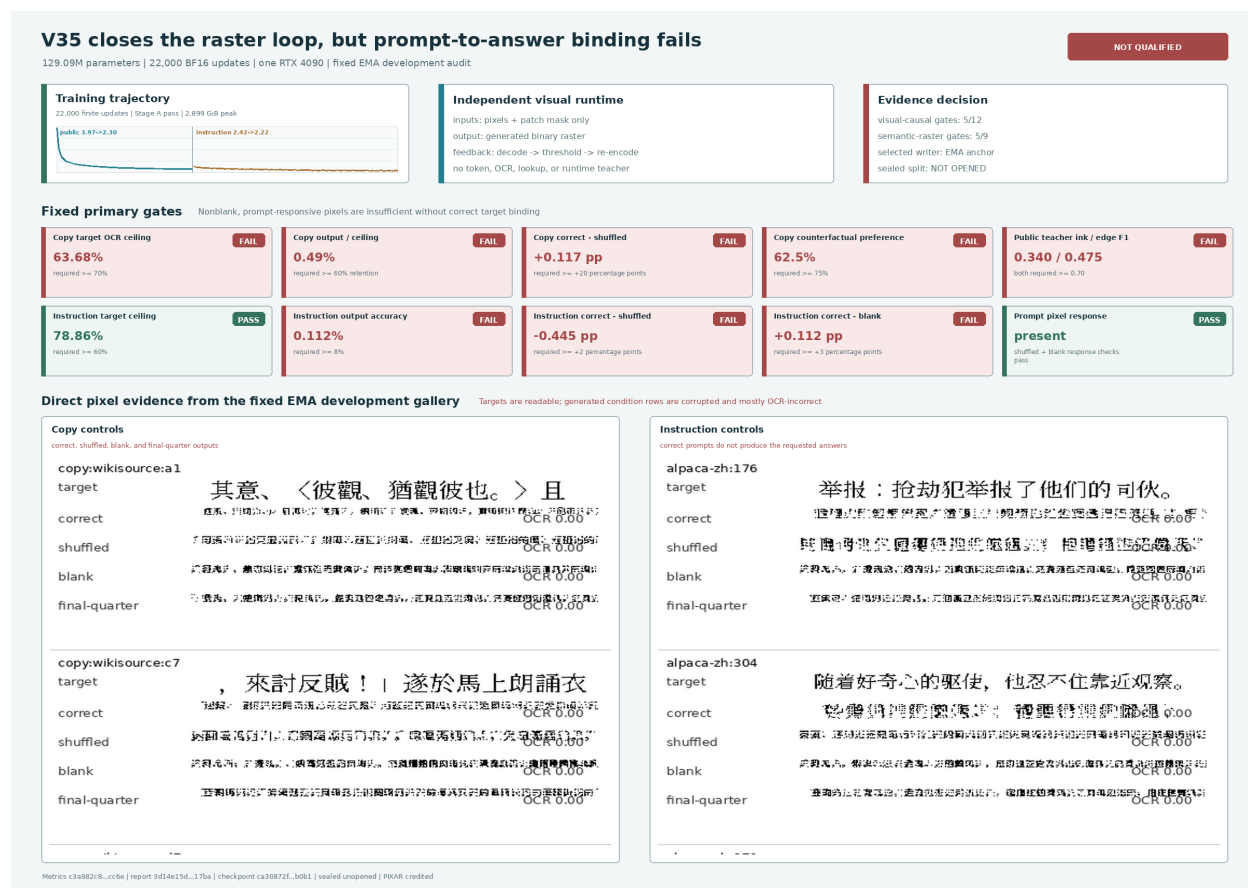


Figure 24: Measured V35 development decision generated from the hash-pinned 22,000-row metric log, report JSON, and immutable EMA galleries. Targets and generated rasters are shown directly; OCR strings never replace model pixels. The model closes the visible feedback loop and responds to prompt interventions, but fails copy, continuation, counterfactual, and instruction-binding gates.

V35 passes 5/12 visual-causal checks and 5/9 semantic-raster checks. Its exact decision is **not-qualified**. The sealed 139 public and 202 instruction records were not rendered, inspected, OCR-scored, or opened. A standalone EMA artifact was exported only under an explicit unqualified diagnostic override; typed-raster and native-image prompts both execute with finite closed-loop states, but their outputs remain incorrect. This is a functioning experimental system and a failed language mechanism. It establishes neither useful autonomous Chinese generation nor parity or compute superiority over a text language model.

The failure localizes the next test. A reconstruction-qualified local glyph latent and a transferred causal transformer can produce prompt-responsive marks without learning answer-level binding. The next model should retain the audited raster boundary while separating a prompt-conditioned multi-patch semantic plan from local raster realization, train that plan with answer-level contrastive and counterfactual objectives, and promote it only if correct prompt interventions move output toward the paired target. External visual or language foundations may be used when they improve this test, provided their provenance, license, training role, and runtime absence or presence remain explicit.

6 Data

The project already has a strong historical Chinese glyph source outside the tracked repo. A local snapshot at `/home/lachlan/ProjectsLFS/incoder/data/historic` contains:

Item	Count
Characters	9,055
Glyph records	84,642
Oracle	32,216
Bronze	23,500
Liushutong	21,923
Seal	6,996

This supports future evidence-bearing historical answer pages but was not used as a character-label channel in the RFLM run. The language trajectory corpus is a 7,017-record manifest built from 16 public-domain Chinese Wikisource book packages. The contribution layer is recorded as CC BY-SA, and the manifest stores source artifacts and rights. Strings exist only in the offline manifest and renderer; model batches contain image tensors.

Eight Noto Sans/Serif CJK fonts provide independent visual views. The run stores the exact font paths, byte counts, and SHA-256 hashes in checkpoints and receipts. Cmap intersection filtering leaves 8,282 supported visible characters; 107 unsupported unique forms account for only 0.0074% of visible corpus instances and are skipped instead of rendered as missing-glyph boxes.

For word-origin and book-layout research, 11 additional local works are registered by SHA-256 in `references/source_books/manifest.json`. They total 963,986,234 bytes and include 9,361 PDF pages plus one EPUB: English and Chinese encyclopedias, an English word-origin reference, and Chinese character- origin and oracle-form references. Relative symlinks avoid duplicating or committing the source binaries. Their rights are unverified, so they are private reference, source-comparison, and evaluation material only; they are not part of the releasable training corpus. PDF/OCR extraction may create offline sidecars, but those strings cannot enter a student batch.

7 Training Objectives

Each 48-fixation page supplies eight sampled causal positions, making the language objective denser than one endpoint per page. The complete training loss is

$$\mathcal{L} = \lambda_f \mathcal{L}_{\text{flow}} + \lambda_e \mathcal{L}_{\text{energy}} + \lambda_r \mathcal{L}_{\text{retina}} + \lambda_v \mathcal{L}_{\text{view}} + \lambda_p \mathcal{L}_{\text{endpoint}} + \lambda_s \mathcal{L}_{\text{stroke}} + \lambda_c \mathcal{L}_{\text{cycle}} + \rho(t) \mathcal{L}_{\text{roll}}.$$

After step 5,200, V7 adds a ramped correction

$$\mathcal{L}_{V7} = \mathcal{L} + \kappa(t) (0.5 \mathcal{L}_{\text{ctx}} + 0.2 \mathcal{L}_{\text{sample}}).$$

The energy term uses 512 cross-render candidate views per batch in the final run. Visual duplicate positives are discovered by image similarity rather than labels. The view term aligns independent renderings in both retinal and candidate-projection spaces. Endpoint and stroke terms improve sparse-ink geometry. The cycle term is weighted toward high-noise examples, where a writer must use recurrent context rather than visible target pixels.

For V6, $\rho(t)$ ramps from zero to one over 400 updates. A rollout subset of eight sequences uses two generated steps, two candidates, and two flow steps per candidate. This follows the distribution-shift lesson of scheduled sampling and DAgger [45, 44]; it remains image-only because

the visited states, selected observations, and recovery targets are pixels and continuous visual fields. The final rollout state cosine reaches 0.961, but selected-image target cosine is only 0.103. This distinction predicts the measured outcome: state recovery becomes robust while sampled visual identity remains weak.

V7 resumes V6 for 800 updates with a 512-form, four-view image-anchor curriculum and a two-step differentiable endpoint on four examples per batch. The anchor retina is refreshed every 200 updates. The selected step 5,800 checkpoint is chosen from preregistered validation checkpoints because it has the strongest combination of independent-anchor context gain (+0.1798) and sampled pixel F1 (0.3636). The continuation adds approximately 261 seconds of training and peaks near 3.0 GiB allocated VRAM on one RTX 4090.

7.1 PVF state objectives

V8–V16 reuse the proven retina but remove the pixel writer from the state ablation. The causal GRU, proposal, and hyperspherical field are trainable. For proposal μ , target image state z^+ , and image-derived candidate states $\{z_j\}$, visual identity uses

$$\mathcal{L}_{\text{proposal-NCE}} = -\log \frac{\sum_{j \in \mathcal{P}} \exp\{s\mu^\top z_j\}}{\sum_j \exp\{s\mu^\top z_j\}},$$

where positives $\mathcal{P}$ are inferred by frozen retinal similarity. A detached last-image condition supplies the calibrated context objective

$$\mathcal{L}_{\text{proposal-ctx}} = \max\{0, m - \ell(z^+ | h_{\text{full}}) + \ell(z^+ | h_{\text{last}})\}.$$

Equivalent identity and context losses are applied to states numerically sampled from the flow. Direct sampled geodesic loss prevents inference-path drift, while contrastive loss prevents collapse to the retinal mean.

V15 has 10,470,273 total parameters, 9,137,345 trainable parameters, and zero classifier or pixel-actuator parameters. It trains in BF16 with 48-image sequences and peaks at 1.181 GiB allocated CUDA memory. The full V14–V15 trajectory records 1,895.24 seconds on one RTX 4090. Step 2,000 was selected on the development image bank before running the frozen evaluator.

V16 resumes selected V15 step 2,000, adds 6,001,536 causal-memory parameters, and resets the optimizer with a new 100-step warmup. It has 16,471,809 total parameters, 15,138,881 trainable parameters, and zero classifier or actuator parameters. The 1,200-update continuation records 2,047.63 seconds, about 92–114 sequences per second, and 1.479 GiB peak allocated CUDA memory. Step 2,200 was selected before the frozen evaluator by maximum development proposal top-1 subject to full-context over last-only, positive normalized context gain, and target cosine above 0.10.

8 Image-First Training and Inference Targets

The next two figures specify the intended full-system contract; they do not claim that the rejected V30 reader can yet generate answer pages. The measured comparison is Figure 19: local spatial alignment changes scores, but exact-suffix assignment remains at chance and writer training stays closed. Presenting the target and the experiment in one figure sequence keeps the research direction visible without turning a concept rendering into evidence.

Training ILM-V: Language as Visual Corpus

Every source becomes a page/glyph image. The proposed student is trained on visual latents, not BPE or word tokens.

CONCEPT · FULL-SYSTEM TARGET

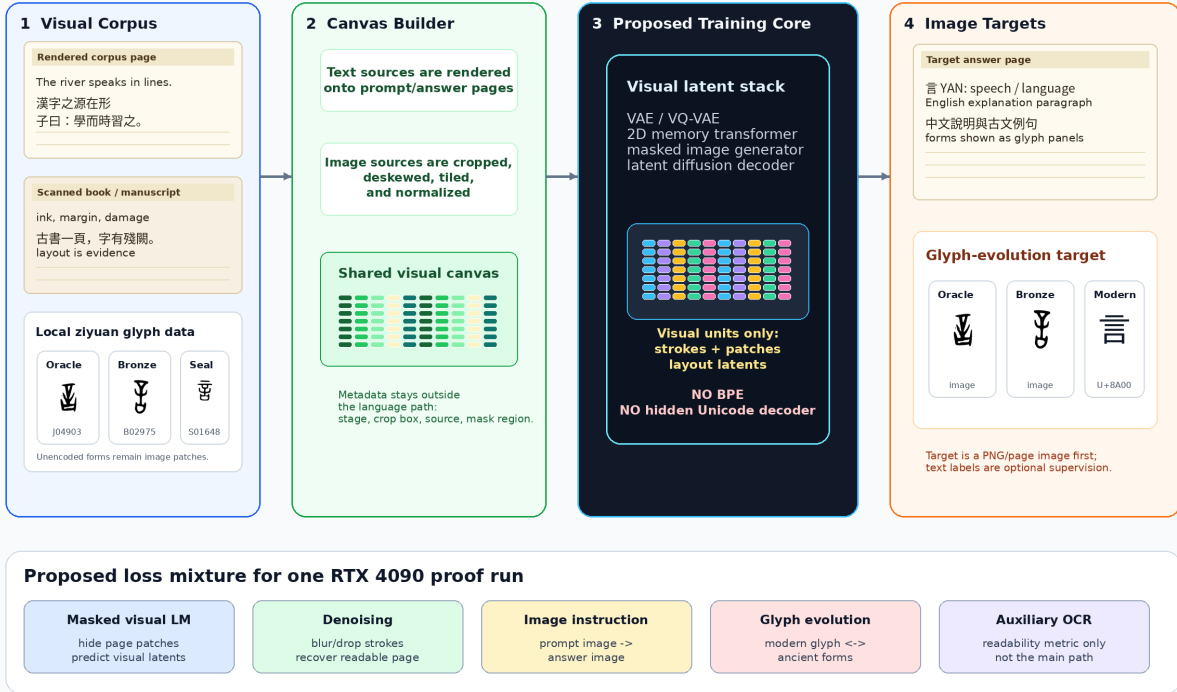


Figure 25: Proposed full-system training target for ILM-V. Rendered text corpora, scanned pages, historical glyph files, and unencoded marks become visual canvases. The intended objective predicts visual latents and images, not BPE tokens. This is a design figure; Figure 19 reports the measured reader experiment.

Figure 25 makes the proposed training contract explicit. Typed corpora are rendered into page images; scanned books and manuscripts are cropped and tiled; historical glyph files become glyph tiles; and marks that are missing from font tables remain valid image patches. Metadata such as crop boxes and stage labels may remain in manifests, but the student path must learn from visual latents. OCR is useful as a readability critic, not as the hidden language channel.

Inference ILM-V: Chat-Like UI, Image-First Output

Typed text and uploaded images become visual prompt canvases. The target student returns a rendered page image first.

CONCEPT · TARGET BEHAVIOR

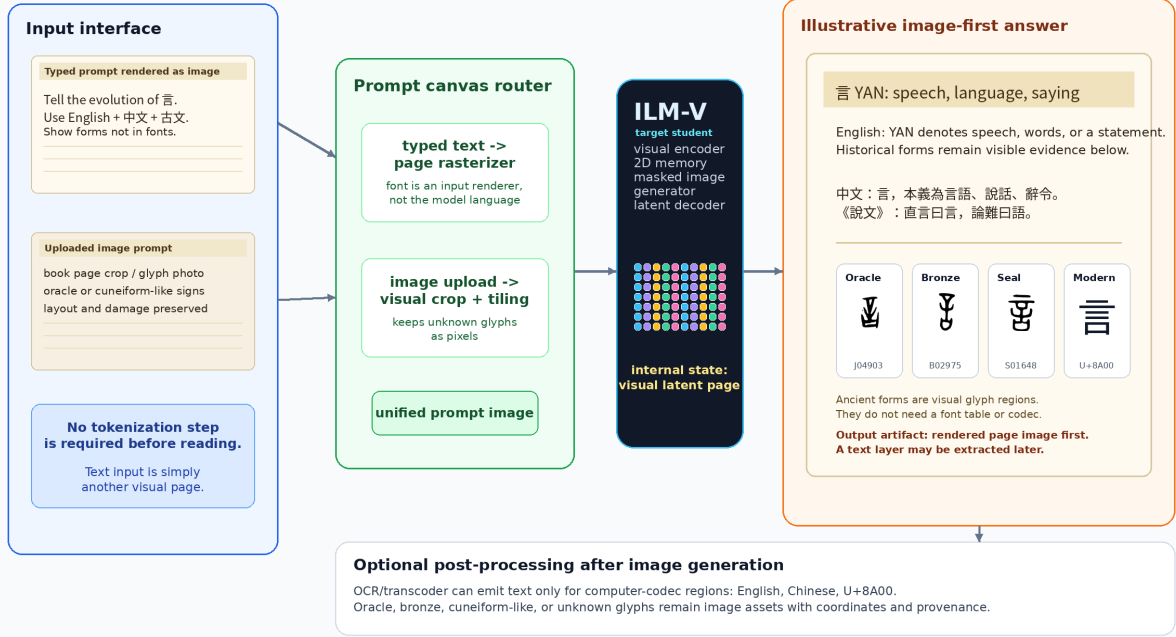


Figure 26: Proposed inference contract for ILM-V. A chat-like interface may accept typed text or uploaded images, but typed text is first rasterized into a prompt image. The target primary output is a rendered page image. OCR or transcoding is optional and applies only to output regions represented by computer codecs. The displayed answer is illustrative, not a V30 output.

Figure 26 shows the target user-facing path. The interface can behave like a normal LLM prompt box. Typed text is rasterized into a clean prompt band; an optional book page, inscription, or glyph crop is normalized and placed beside or below it on the same canvas. Borders, whitespace, and coordinates make the relation visible rather than encoding it in hidden IDs. A successful ILM should answer a typed question alone or a grounded question such as “What does the marked form mean, and how did it evolve?” about an attached page. It returns a page image containing modern language and historical glyph regions only after the corresponding visual-language gates pass. A later OCR/transcoder may attach a searchable layer for encodable spans; ancient or unencoded glyphs remain image regions with coordinates and provenance.

The formal contract is:

$$\begin{aligned}
 r_{\alpha}(\text{text prompt}) &\rightarrow x_{\text{text}}, \\
 g_{\beta}(\text{uploaded page or glyph}) &\rightarrow x_{\text{source}}, \\
 \text{Compose}(x_{\text{text}}, x_{\text{source}}) &\rightarrow x_{\text{prompt}}, \\
 f_{\theta}(x_{\text{prompt}}) &\rightarrow y_{\text{page}}, \\
 o_{\psi}(y_{\text{page}}) &\rightarrow \text{optional text layer} + \text{glyph-region metadata}.
 \end{aligned}$$

Here r_{α} and Compose are deterministic UI adapters, not language components. Either visual region may be absent. The core f_{θ} receives the resulting pixels and maps them to answer image y_{page} . The optional reader o_{ψ} may recover coded text where possible, but unencoded historical forms remain

Single-4090 ILM: Minimal Working Path

One visual substrate from ordinary prompts and book pages to generated answer pages

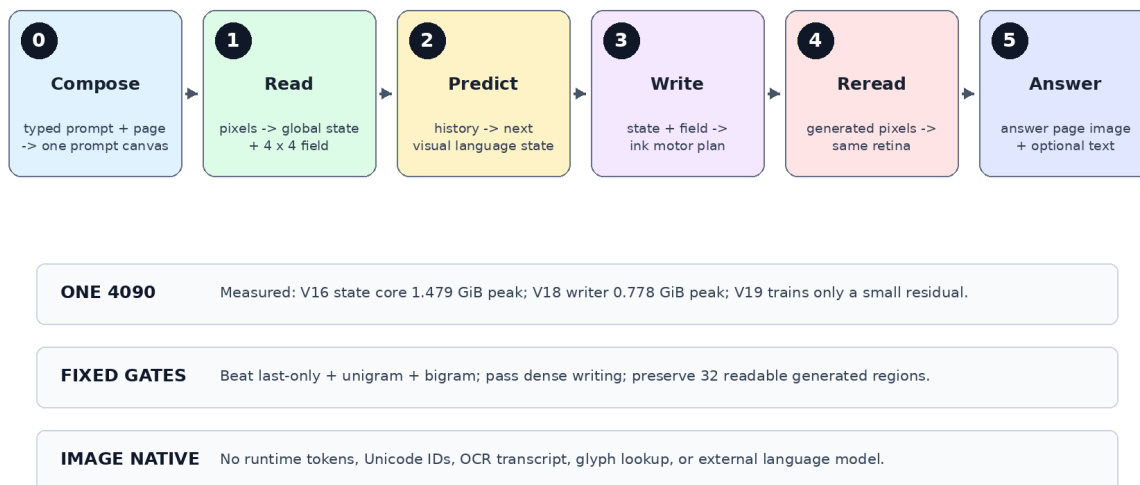


Figure 27: A staged curriculum that keeps each causal proof small enough for one RTX 4090 before any page-scale expansion.

valid image regions.

A useful answer canvas should resemble a page excerpt from a book or corpus:

- a title region rendered as text image,
- one or more modern-language paragraphs, for example English and Chinese,
- a classical-style line or quotation when appropriate,
- glyph panels for forms that may not exist in computer fonts,
- a provenance strip with source IDs or nearest retrieved exemplars.

This schema is intentionally different from “text answer plus illustrations”. The generated page image is the primary output object, and the text layer is secondary.

9 Single-4090 Curriculum

Figure 27 records the broad curriculum. The executable path is intentionally simpler than the complete research space:

1. Read 32×32 writing images into one global state and one small 4×4 spatial field. V16 already provides the retina and causal state.
2. Predict the next continuous visual state from image history and require it to beat last-only, unigram, and bigram controls.

3. Draw that state with a continuity-preserving local route. V19 shows that a small optional residual is ignored; V20 makes field detail causal; V21 makes both occupancy and detail field-causal but rejects disjoint patch rasterization. The successor may overlap only bounded neighboring patches.
4. Render typed prompts and optional source pages into visual frames, then predict local/global answer fields from open question–answer trajectories. Require held-out prompt counterfactuals and prompt-shuffle controls.
5. Feed generated answer pixels back through the same retina and require a readable, prompt-dependent multi-frame autonomous rollout.
6. Add model width or a second GPU only after the preceding fixed gate passes; use the second GPU for throughput or independent replication, not to hide an unresolved objective.

This uses one retina, one causal field, one deterministic motor planner, and one write–reread loop. Stochastic surface generation, billion-parameter scaling, and broad multimodal alignment are deferred until this minimal system works. The same Visual Language Stream can later expose geometric depth and time, but 3D glyph strings and character movies are also deferred until the 2D write–reread gate passes.

The 16.47M PVF V16 state model, 5.73M V17 actuator, and 2.36M V18 motor planner are deliberately small enough for one RTX 4090. Their measured peaks are 1.479, 1.588, and 0.778 GiB, respectively, leaving substantial capacity headroom for controlled corrections. V20 reduces the causal routing test to two exactly matched 0.506M-parameter arms, peaking at 0.323 and 0.397 GiB, and cheaply rejects the writer despite positive field causality. This is not an end-to-end efficiency victory: the causal language core remains below a symbolic bigram, dense topology remains imperfect, and autonomous continuation has not passed. “More efficient than an LLM” remains an unsupported hypothesis until quality is matched on a named task.

10 Target Demonstration: Visual Word-Origin Book

The first demo should answer either a normal rendered prompt such as “Explain the evolution of the Chinese character yan” or the same question composed with a photographed book page. It must also support English word-origin questions. The target answer image should contain:

- a title and concise modern explanation,
- English and Chinese paragraphs rendered as page text,
- a classical-style line such as “yan is the visible trace of speech”,
- a historical timeline with real oracle, bronze, seal, and modern glyph exemplars,
- stage labels and citations as rendered image text.

The existing Zhong figure in Figure 28 is a compact stage-timeline example; the Yan figures in Figures 1 and 26 show the fuller answer-page style. Together they define a future benchmark for combining prompt semantics, visual page composition, and historical-script generation; they are not measured V35 outputs.

Example Output: Evolution of Chinese Character Zhong

A target answer is itself an image: explanation plus historical forms.



Figure uses local hanziyuan-derived glyph files when available; generation targets should retrieve/cite real exemplars.

Figure 28: Target output style for a glyph-evolution question. The answer is not merely text: it is a readable image with historical visual forms. The example uses local hanziyuan-derived glyph files when available.

11 AgInTi Artifact Loop

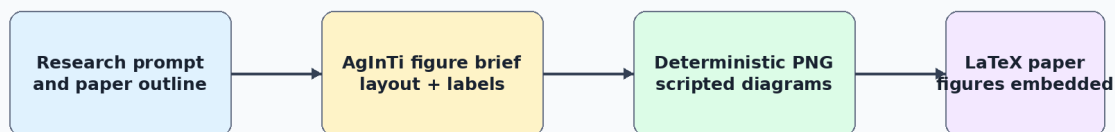
The figure brief for AgInTi-style refinement is stored in the local AgInTi prompts subfolder. The current paper uses scripted PNG generation to keep the artifact reproducible.

12 Evaluation

The model should be evaluated as both a language model and a visual generator:

- Fixed-bank top-1, top-5, and reciprocal rank for full context and a last-fixation ablation.
- Random-weight, chance, unigram, and symbolic bigram baselines on identical contexts.
- Cross-font retina-oracle retrieval, separated from recurrent language prediction.
- Generated-image target similarity, pixel F1, ink occupancy, variance, and energy-reranked hit rate.
- Autonomous rollout length, throughput, VRAM, and blinded human readability.
- Student-boundary audit for token IDs, Unicode IDs, OCR, codebooks, labels, and external models.
- Visual Word-Origin Book tasks: factual proposition accuracy, modern-text readability, prompt-image grounding, attested glyph-stage and source accuracy, unencoded-region preservation, and explicit synthetic-form labeling.
- Optional output transcription scored separately from the native answer image so an OCR sidecar cannot conceal a failed visual answer.

AgInTi-Assisted Research Artifact Loop



Why scripted PNGs now?

They are reproducible, editable, and compile cleanly in the paper while leaving room for later AgInTi/image-model refinement

Figure 29: AgInTi-assisted artifact loop. The current figures are deterministic scripted PNGs so the paper is reproducible; the figure brief can later be sent to an interactive AgInTi/image agent for visual refinement.

System	Measured signal	Interpretation
Whole-page flow	velocity MSE 0.851	Unreadable page texture; negative baseline
Global visual memory	top-1 2.23% over 2,016	Above random, but 0/16 historical paraphrases
Causal InkStream	validation ink F1 0.639	Autonomous repeated-grey collapse; negative baseline
RFLM V5 retina oracle	top-1 97.65%	Strong cross-font visual alphabet
RFLM V5 recurrent state	top-1 0.91%	Below last-only 1.36%, unigram 1.86%, and bigram 13.58%
RFLM V6 recurrent state	top-1 1.20%	Improved, but below last-only 1.69% and both language baselines
RFLM V6 pixel writer	context gain +0.0077	Positive but weaker than V5's +0.0211
RFLM V6 autonomous loop	ink ratio 1.168	Stable character-scale marks remain unreadable
RFLM V7 recurrent state	top-1 2.31%	Above last-only 2.02% and unigram 1.86%, far below bigram 13.58%
RFLM V7 calibration	log-probability gain -0.2155	Much better than V6's -0.9066, but full history still lowers mean target probability
RFLM V7 pixel writer	context cosine +0.0303	Passes generated-signal gate; autonomous writing remains unreadable

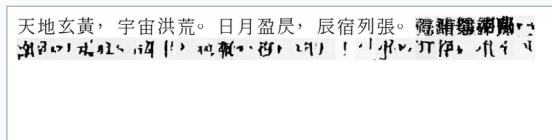
The fixed evaluation contains 512 common Han candidates with four independent font views

Closed-loop visual trajectory training: what changed

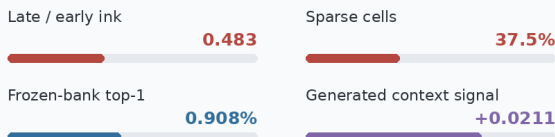
Matched 32-cell autonomous generation; same prompt, seed, sampler, and 11.69M-parameter model

V5 Clean-prefix training

Ink thins under its own visual feedback

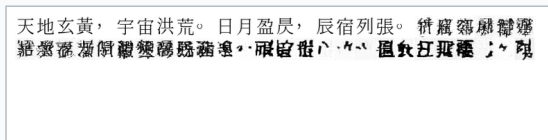


Rendered prompt followed by 32 model-generated image cells

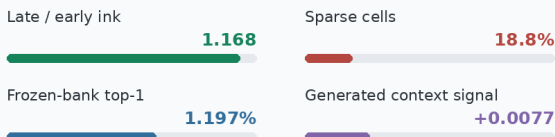


V6 Model-induced rollout training

Ink survives, but the writing remains unreadable



Rendered prompt followed by 32 model-generated image cells



Measured verdict closed-loop stability improved; contextual language and readable generation did not

Frozen glyph bank SHA-256 543987ddd754135cff34dbee0f6f91ce1971741d0807754c65650031121181c4

Figure 30: Matched autonomous evidence for V5 and V6. The prompt, seed, sampler, model size, and 32-cell horizon are fixed. Model-induced rollout training prevents the earlier loss of ink and halves the sparse-cell fraction, but neither continuation is readable and generated target signal weakens.

and 2,423 eligible held-out contexts. V6 and V7 use the identical bank SHA-256. Selected V7 raises full-context top-1 from 1.197% to 2.311% and top-5 from 3.219% to 5.613%, while the retina oracle remains near 98.27%. Full context now ranks above V7 last-only (2.022%) and unigram (1.857%) but remains far below symbolic bigram (13.578%). Its normalized target-log- probability gain is -0.2155 , improved from a corrected V6 value of -0.9066 but still negative. Raw target-score gain stays positive in both models, showing that raw energy was a misleading metric. The formal acceptance result remains false, and retina-oracle accuracy cannot be cited as language accuracy.

PVF uses a separate frozen evaluation with the same 512 objects and four views for V14 through V16, bank SHA-256 543987dd...1181c4, and 1,788 eligible held-out contexts. V15 proposal top-1 is 5.872%, compared with 4.418% last-only, 1.734% unigram, 0.168% random dynamics, and 13.143% symbolic bigram. Proposal top-5 is 12.192%, and normalized context gain is $+0.07069$. The stochastic field obtains 3.412% top-1, 7.662% top-5, $+0.03032$ normalized context gain, and $+0.08053$ sampled context cosine gain. Random-dynamics sampled target cosine is 0.03180, versus 0.21576 for V15. The retina oracle is 98.546%.

V16 was selected at step 2,200 without reading that frozen bank and evaluated once. Its proposal obtains 6.264% top-1 (112/1,788), versus 3.971% last-only, 1.734% unigram, 0.224% random dynamics, 0.195% chance, and 13.143% symbolic bigram. Its proposal context gain is $+0.07732$. The stochastic field obtains 3.691% top-1 versus 3.244% last-only, normalized context gain $+0.03579$, sampled target cosine 0.19506, and sampled context cosine gain $+0.07947$. V16 adds seven correct proposal contexts over V15, while proposal top-5 and target cosine decline. This is directional evidence and not a statistically established memory advantage; paired predictions on a new frozen bank are required for attribution.

Thus the proposal and flow state gates are accepted, while the bigram language gate remains

V7 Anchor Identity: Better Signal, Still Not Language

Same prompt, seed, model size, sampler, and 32-cell horizon

V6 closed loop

天地玄黃，宇宙洪荒。日月盈昃，辰宿列張。攸嘉謬鼎鬱鬱
維繫微微爾爾變變隨隨通通。祇留世々々々風數狂舞擾々々々

top-1 1.20% normalized context -0.907 generated signal +0.0077

ink ratio 1.168 sparse 18.75% unreadable

V7 anchor + sampled identity

天地玄黃，宇宙洪荒。日月盈昃，辰宿列張。攸嘉謬鼎鬱鬱
維繫微微爾爾變變隨隨通通。祇留世々々々風數狂舞擾々々々

top-1 2.31% normalized context -0.215 generated signal +0.0303

ink ratio 1.050 sparse 15.63% unreadable

Verdict: V7 closes most of the calibrated context deficit, but dense pseudo-glyphs are not readable Chinese. Separate visual-state prediction from rendering.

Figure 31: Matched autonomous evidence for V6 and selected V7 step 5,800. V7 improves calibrated retrieval and generated target signal while retaining stable ink, but visual inspection still finds dense unreadable pseudo-glyphs.

false. Separately, V17 accepts causal visual actuation and rejects readable topology. V18 then passes every automatic development topology gate and makes many held-out forms readable, but leaves dense-form and prospective human-readability gates unresolved. V19 then rejects an additive local-field repair: dense F1 is 0.7278, but field shuffling reduces it by only 0.0088 and field removal by only 0.0054. V20 then forces unavailable within-block detail through the field. Correct dense F1 exceeds shuffled-field by 0.1218 and zero-field by 0.3613, with exact quadrant locality, so the local-causality hypothesis passes. Writer selection still fails, and endpoint dense advantage over the equal-capacity control is only 0.0179 against a fixed 0.03 margin. V21 removes the remaining global spatial route. At its best diagnostic snapshot, correct dense F1 exceeds shuffled-field by 0.1739 and zero-field by 0.3465, identity top-1 is 79.10%, target cosine is 0.8331, locality is 1.0, and every exact-basis invariant passes. The tiled-global control collapses to repeated patches. Overall, simple, and medium F1 remain below their fixed gates, so no candidate selects and no paired, human, or frozen stage is authorized. V22 then tests the first prompt-to-answer image stream. Its candidate changes answer identity correctly for only 0.78% of query-counterfactual pairs, versus 0% for the query-blind control, while identity and writer quality are nearly equal. Every candidate selector argmax lands on the operation image. Thus the implemented stream boundary works, but an entropy-sharpened single-frame selector cannot perform the required multi-frame relation. V23 then qualifies and freezes an image canonicalizer before learning the relation. Its candidate reaches 99.80% minimum query/operation switch on a fresh paired audit; the corresponding blind controls are exactly invariant. The opaque visual review and single frozen evaluation also pass, with 99.46% frozen identity top-1 and 0.7848 pixel F1. This turns V22's interface into the first accepted bounded image-prompt-to-image-answer circuit in the project. V24 then replaces six absolute frame roles with visible headers in a 15–24-frame packet stream and closes a two-frame visual autoregressive loop. The fresh candidate reaches 99.22% query, 99.32% operation, and 99.61% generated-history switch accuracy; each corresponding blind control is exactly invariant.

PVF branch	Frozen signal	Interpretation
V14 proposal	top-1 2.52%	First proposal gate pass; above last-only 2.24% and unigram 1.73%
V14 state flow	top-1 2.35%	First stochastic state gate pass
V15 proposal	top-1 5.87%	Above last-only 4.42%, unigram 1.73%, and random 0.17%; context gain +0.0707
V15 state flow	top-1 3.41%	Above last-only 2.68% and unigram; sampled context cosine gain +0.0805
V16 proposal	top-1 6.26%	112/1,788; above last-only 3.97% and V15 by seven contexts; context gain +0.0773
V16 state flow	top-1 3.69%	Above last-only 3.24% and unigram; sampled context cosine gain +0.0795

Table 24: Frozen Predictive Visual Field results. V15 and V16 branches pass their visual-state gates but remain below the 13.14% symbolic bigram.

V17 frozen actuator metric	Correct state	Shuffled state	Gate
Global identity top-1	58.594%	0.977%	pass
Target cosine	0.71301	0.08612	pass
Pixel F1	0.43849	0.28001	fail (< 0.50)
Ink fraction	0.15947	0.15967	controlled
Human readability	rejected	rejected	fail

Table 25: Untouched V17 evaluation with identical style and noise in each paired branch. Causal visual identity succeeds while readable topology fails.

The opaque audit scores 47/48 and 48/48 for the two output frames. On the single frozen run, frame-1 unseen-identity accuracy is 98.67%, frame-2 label accuracy is 99.73%, and held-out-length scores are 98.47% and 100%. Every fixed gate passes. V25 then removes the packet grammar and exposes the larger gap. Full natural- Chinese visual history reaches 1.123% top-1, above 0.146% last-only and 0.342% shuffled history, but below 1.611% image unigram and 12.158% symbolic bigram. Six fixed gates reject the language model. The exploratory writer’s nonblank images preserve density while obtaining zero identity top-1, so appearance stability cannot substitute for semantic continuation. The frozen partition remains sealed. V26 then separates last-glyph appearance from the preceding visual history and replaces one-point prediction with continuous future particles. It passes the full-versus-last log-probability gate, the perfect retina-oracle gate, the image-only boundary, and the resource gate. It fails the more specific causal tests: full history gains only 0.0197 nat over suffix-4 and 0.00369 nat over a suffix-preserving shuffle. Pixel-identical suffix pairs produce a 4.628 mean history-residual difference but exactly 50% correct target ranking. Thus the representation reacts to history while the predictive distribution does not use that reaction correctly. V26 rejects the proposed factorization and loss; the frozen partition remains sealed and no writer is trained. A post-hoc frozen-state probe further shows that this is not solely a particle- proposal bottleneck: a deterministic nonlinear scorer reaches 50.684% from history and 50.342% from fused state while a raw-retina identity control reaches 99.951%. This motivates jointly learning visual context representation and candidate compatibility rather than preserving V26’s encoder unchanged. V27 performs that joint test. Its full-context pair assignment is 50.708%, versus 50.562% after shuffling the earlier prefix, while natural top-1 is 1.611%, below 2.051% image unigram and 12.598% symbolic bigram. The raw retina still gives 99.951% cross-font identity

V18 development motor-plan metric	Correct intent	Shuffled intent	Gate
Global identity top-1	73.633%	0.977%	pass
Target cosine	0.84618	0.07162	pass
Pixel F1	0.65772	0.31290	pass (> 0.60)
Ink fraction	0.17505	0.17505	controlled
Human readability	simple/medium clear	unrelated	dense forms fail
Frozen images instantiated	no		sealed

Table 26: Fresh V18 development audit with style held fixed and only continuous intent shuffled. Automatic topology gates pass, but frozen promotion is withheld because the human rubric was not numerically prespecified.

and candidate permutation error is exactly zero in the two-candidate pair control. Separately, learned 1,024-bank identity is 94.873% and misses its 99% gate; the two retrieval sizes are not directly comparable. V27 therefore rejects the global query/key compatibility route; its frozen partition remains sealed. V28 then freezes the raw retina and distributes supervision across 16 causal positions and three future horizons. The same-scope 1,024-way EMA semantic identity reaches 96.436%, 4.395 percentage points above the raw-retina route, and full target log probability gains 0.215 nat over the shuffled-prefix arm. Nevertheless, natural top-1 is 1.416%, below 1.855% unigram and 13.135% bigram, while full-context pair assignment is 49.561% versus 49.951% after prefix shuffling. The pair mean-margin gate passes, but the rank gates fail. V28 therefore rejects dense global future energy as a sufficient binding mechanism; its frozen partition remains sealed and no writer is trained. V29 then makes each candidate image cross-attend to retained per-cell context states and trains the centered full-minus-suffix increment directly. Full target log probability gains 3.102 nat over suffix-preserving prefix shuffle, but natural top-1 is only 2.344% versus 13.867% for bigram. Exact-suffix incremental assignment is 50.708%, versus 49.561% shuffled, and both-correct is 8.984%. Raw two-candidate identity is 99.951%, frozen-semantic 1,024-way identity is 96.436%, and all equality and permutation controls pass. V29 thus separates a strong global order effect from candidate-specific binding. It also shows algebraically and empirically that a shared suffix baseline cancels from the aggregate two-by-two assignment margin. The critic itself must learn the missing interaction; changing only the baseline cannot supply it. V30 makes that context state predict a candidate-independent 4×4 retinal field and tests aligned local scoring against a parameter-identical tiled-global control. Patch reversal changes spatial scores, but natural top-1 is 1.270% versus 2.490% for the global route, and exact-suffix assignment is 50.049% versus 50.586%. The spatial route is therefore sensitive to local geometry without learning a useful conditional next-image distribution. The combined result establishes that a small image-only student can learn causal language signal, that a small continuous field can control topographic generated ink, and that several visual roles can be composed into a rendered answer stream whose second frame depends on rereading the first. It does not establish useful autonomous language. V25 shows that simply replacing the hand-authored grammar with a generic causal visual field is insufficient. V26 shows that separating appearance from history is also insufficient when the objective permits changed residuals with nearly unchanged target preference. V27 shows that joint dot-product candidate compatibility is also insufficient: it preserves raw target visibility in the pair control but neither passes the separate learned full-bank identity gate nor makes earlier order useful. V28 shows that freezing that retina, improving semantic identity, and adding dense ordered future supervision can make earlier order affect probability and margin without producing correct binding. V29 shows that candidate-conditioned cross-attention and explicit prefix-increment contrast can amplify the

natural order effect while exact-suffix assignment remains at chance. V30 shows that one aligned deterministic next-image field is also insufficient and is weaker than its matched global route. V31 then models a coherent multimodal continuous field distribution. Its spatial route detects order and local permutation and produces diverse context-dependent fields, yet natural path top-1 is 0.098% and exact-suffix path and autonomous assignment remain at 50.488% and 50.195%. Thus adding continuous transport to the same one-glyph target does not repair language binding. The next experiment should stop optimizing candidate-bank retrieval and instead directly generate a multi-glyph answer raster from a causal visual state. It must retain suffix-preserving shuffle, visual-frequency, symbolic sequence, content-swap, prompt-counterfactual, and generated-history controls. Only after held-out semantic compositions and readable direct pixels pass should variable page length, multiple generated lines, or raw-book scale expand the task. This order is also consistent with measured error accumulation in iterative diffusion chains [46].

13 Conclusion

Visible writing is a legitimate modeling substrate, but replacing symbolic tokens with pixels does not make language learning easy or automatically more efficient. The experimental sequence separates several requirements that are often conflated: preserving glyph form, using ordered visual context, binding a prompt relation, writing pixels, and rereading generated output. V23 and V24 establish bounded visual relation following and generated-pixel causality under fixed grammars. V25–V31 then show repeatedly that context-sensitive continuous states can fail to select the correct natural-language future.

V34 resolves one interface bottleneck. Its 7.42M-parameter continuous codec preserves held-font modern Chinese and held-family historical forms with about 0.998 ink and edge F1, without a discrete visual vocabulary, and trains in 321 seconds with 2.38 GiB peak allocated memory. This is a strong visual representation and reconstruction result. It is not evidence that the latent organizes meaning, grammar, or likely futures.

V35 tests the stronger closed-loop hypothesis using that codec and an explicitly credited PIXAR initialization. The 129.09M-parameter system completes 22,000 finite BF16 updates in 2.77 hours with 2.899 GiB peak allocated memory on one RTX 4090. Its packaged inference path receives pixels and a visual-position mask, emits binary raster patches, and feeds only decoded, thresholded, and re-encoded visible patches into later context. It contains no tokenizer, Unicode or character IDs, OCR, retrieval, codebook, lookup table, or runtime teacher. This makes V35 a complete and reproducible raster-input/raster-output experimental runtime.

The language result is nevertheless negative. Autonomous copy character accuracy is 0.3125%, correct copy prompts exceed shuffled prompts by only 0.117 percentage point, and copy target preference is 62.5%. Autonomous instruction accuracy is 0.1116%, lower than the shuffled-prompt condition. Generated strips are finite, nonblank, and responsive to pixel interventions, but their marks are corrupted and not bound to the requested target. V35 passes 5/12 visual-causal gates and 5/9 semantic-raster gates and is **not-qualified**. The sealed split remains unopened. Training loss, nonblank pixels, and nonempty OCR are not substitutes for counterfactual semantic evidence.

The strongest supported claim is therefore narrow: one consumer GPU is enough to build, train, audit, package, and execute a self-contained continuous visual language experiment at 129M parameters, but the tested objective does not learn useful autonomous Chinese language. No Qwen, GPT, or general-LLM parity is shown, and no matched evidence supports compute superiority over token models. The runtime is independent of external services, not independent of provenance: PIXAR supplies external causal initialization, V34 supplies the project codec, and the unstated

PIXAR weight license prevents redistribution of the derived artifact.

The next bounded experiment should not merely enlarge V35. It should learn an answer-level, multi-patch visual semantic plan with strong paired, counterfactual, and contrastive supervision, then condition a separate raster renderer on that plan. Representation-rich external visual or language foundations are acceptable if they make the experiment work and if source, license, training role, and deployed presence are explicit. Promotion still requires that correct prompts move autonomous pixels toward the correct paired answer more than shuffled, blank, partial, and paraphrased controls.

A Visual Word-Origin Book remains the product target: rendered questions should produce readable answer images containing modern explanation and provenance-linked historical forms, including forms absent from computer encodings. V35 does not yet answer such questions. Page-scale reading, 3D writing, motion, speech, and broad multilingual generation remain future extensions of the same sensory interface and require separate evidence.

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